International Journal of Machine Tools & Manufacture 202 (2024) 104211

Contents lists available at ScienceDirect

International Journal of Machine Tools and Manufacture

journal homepage: www.elsevier.com/locate/ijmactool

Research Article

Optimal milling cutter helix selection for period doubling chatter suppression

M. Sanz-Calle [a,b,*], A. Iglesias [a], L.N. López de Lacalle [b], Z. Dombovari [c], J. Munoa [a,b]

[a] Dynamics & Control, IDEKO, Arriaga industrialdea 2, 20870 Elgoibar, Spain
[b] Mechanical Engineering Department, Faculty of Engineering of Bilbao, University of the Basque Country, 3 Rafael Moreno Ave., 48013 Bilbao, Spain
[c] MTA-BME Lendület Machine Tool Vibration Research Group, Department of Applied Mechanics, Budapest University of Technology and Economics, Muegyetem rkp. 3, H1111 Budapest, Hungary

ARTICLE INFO

MSC:
00-01
99-00

Keywords:
Milling
Chatter
Interrupted cutting
Period doubling
Helical tool

ABSTRACT

In high speed milling, interrupted cutting conditions can lead to period doubling chatter vibrations. While many studies have already confirmed that the use of helical tools can effectively shrink or remove these regions of unstable cutting, none of them has provided clear guidance to select the minimum helix that completely cancels the period doubling lobes. This study addresses this gap by introducing a novel analytical formula for a critical tool helix pitch: if the helix pitch is below the critical flip depth of cut of the straight helix cutter multiplied by π, the flip lobes will totally vanish. This rule is not only valuable for chatter-free process planning purposes, but it also establishes exact limit below which the fast and simple zeroth order stability algorithm can provide exact stability boundaries for helical tools. The effectiveness of the formula is numerically corroborated over three different milling scenarios: thin wall milling, slender tool and machine tool structure chatter cases. Finally, the findings are validated through experimental cutting tests.

1. Introduction

Technological advancements in recent decades in cutting tool materials and the availability of machines with heads that offer a wider range of speed and power have made high speed milling possible for multiple materials, including steel, aluminum, titanium, and composites. When operating under these high speed conditions, it is common for the excitation frequencies of the process to fall within the range of the flexible modes of the machine, head, tool, or workpiece. Under these conditions, the material removal rate can be limited by the occurrence of two different types of chatter: Hopf chatter and period doubling chatter. Even though both are caused by the chip thickness regeneration effect, they are associated to different origins.

Hopf chatter is produced by the 'mean' dynamic cutting force and therefore can occur regardless of how interrupted the cutting is. The suppression of Hopf chatter can be tackled in multiple manners, as reviewed by Munoa [1]. Special tool geometries, increasing the stiffness and damping of the system and spindle speed variation techniques can be used and combined to achieve stable cutting. The prediction of the spindle speed — depth of cut Hopf stability limits can be conducted using the traditional zeroth order algorithm (ZOA) introduced by [2]. This algorithm offers a rapid and analytical frequency domain solution by disregarding the intermittent nature of the milling force. However, it does not consider the deviations in Hopf boundaries due to the mode interaction effect caused by the interrupted cutting as reported in [3].

In contrast, period doubling chatter is induced by time-periodic fluctuations of the dynamic cutting force and occurs when the cutting is interrupted – i.e. low radial immersion and low number of teeth – [4]. Period doubling chatter generates additional unstable lobes at spindle speeds where any of the harmonics is twice the natural frequency of the limiting mode. As shown in Fig. 1a, these additional lobes further limit the combinations of spindle speeds and axial depths of cut that provide stable (chatter-free) machining. Because harmonics of the dynamic cutting force lose amplitude as higher order of the harmonic is considered, period doubling chatter is more prone to happen in high speed milling, where the tooth passing frequency can be comparable or higher than the natural frequency of the limiting modes. For example, in high speed aluminum milling, cutting tools with low number of teeth working at very high spindle speeds can result in strong harmonics at frequencies near or above the modes of the tool or the part, increasing the flip chatter likelihood.

On the one hand, predicting stability boundaries in interrupted processes where period doubling unstable areas are present necessitates methods capable of capturing variations in cutting forces. The multifrequency algorithm originally introduced in [5] can describe the Hopf and flip lobes simultaneously by considering a discrete number of harmonics of the dynamic milling force. Alternatively, a semi-analytical approach can compute Hopf and flip lobes separately for simple straight

* Correspondence to: Arriaga industrialdea 2, 20870 Elgoibar, Spain.
E-mail address: msanz@ideko.es (M. Sanz-Calle).

https://doi.org/10.1016/j.ijmachtools.2024.104211
Received 17 June 2024; Received in revised form 5 September 2024; Accepted 7 September 2024
Available online 10 September 2024
0890-6955/© 2024 The Authors. Published by Elsevier Ltd. This is an open access article under the CC BY-NC-ND license (http://creativecommons.org/licenses/by-nc-nd/4.0/).

Nomenclature

β	Straight helix harmonic of the directional factor
δ	Helix factor
η	Tool helix angle (deg)
$\eta_{r,\mathrm{crit}}$	Critical helix angle for flip suppression (m)
Λ	Eigenvalue
$\lambda_{r,\mathrm{min}}$	Dimensionless flip frequency at the minimum
i	Imaginary unit
Ω	Spindle speed (rpm)
ω	Frequency (rad/s)
ω_Z	Tooth passing frequency (rad/s)
$\omega_{\mathrm{c,min}}$	Chatter frequency at the minimum (rad/s)
ω_{c}	Chatter frequency (rad/s)
ω_{n}	Natural frequency (rad/s)
$\boldsymbol{\Phi}$	Receptance (m/N)
$\boldsymbol{\Phi}_r$	Receptance evaluated at modulation r (m/N)
τ	Regenerative delay (s)
θ	Mode direction angle (deg)
φ	Angular position (rad)
φ_0	Angular position of the first flute at the tip (rad)
φ_i	Angular position distribution of the flute i (rad)
φ_η	Helix lag angle (rad)
φ_{en}	Engagement entry angle (rad)
φ_{ex}	Engagement exit angle (rad)
φ_{p}	Cutter pitch angle (rad)
ζ	Relative damping
A	Temporal directional factor
a	Axial depth of cut (m)
A_r	Harmonic of the directional factor
$a_{r,\mathrm{min},\eta}$	Critical depth of cut of flip lobe with helical tool (m)
$a_{r,\mathrm{min}}$	Critical depth of cut of flip lobe with straight helix (m)
B	Temporal directional factor of differential axial disk
b	Contribution of each tooth to directional factor
D	Tool diameter (m)
F	Dynamic milling force (N)
F_r	Harmonic of the dynamic milling force with straight helix (N)
f_Z	Feed per tooth (m)
$F_{r,\eta}$	Harmonic of the dynamic milling force (N)
g	Screen function
k	Reflected stiffness (N/m)
k_r	Cutting force ratio
K_{rc}	Radial cutting coefficient (N/m)
K_{tc}	Tangential cutting coefficient (N/m^2)
p	Tool helix pitch (m)
$p_{r,\mathrm{crit}}$	Critical helix pitch for flip suppression (m)
q	Dynamic modal displacement (m)
q_r	Dynamic modal displacement harmonic (m)
r	Number of harmonic/flip lobe order
$r_{\beta,r}$	Straight helix harmonic ratio
t	Time (s)
T_Z	Tooth passing period (s)
Z	Number of teeth
z	Height from tool tip (m)

fluted cutters, enhancing computational efficiency [6]. Discrete-time methods like semidiscretisation (SDM, [7]) or full discretisation [8] can be used to construct the stability charts for interrupted milling cases featuring complex cutter geometries with high computational efficiency. However, as shown in [6], the calculation time is several orders above the classic ZOA solution.

On the other hand, the methods for suppressing period doubling chatter are generally different from those traditionally applied to Hopf chatter and have received less attention in the literature. Because the origin of period doubling chatter is associated to interrupted cutting, smoothing the dynamic milling forces is by far the most effective way to decrease the period doubling chatter likelihood. Consequently, the inclusion of helical flutes in the milling tool greatly decreases the period doubling chatter likelihood. The gradual entry of the flute into the material smoothens the cutting force and decreases the stationary vibration level as demonstrated in [9], the effect on the period doubling lobes was not studied, though. Alternatively, spindle speed variation has been numerically and experimentally confirmed to be able to reduce or even eliminate the flip chatter likelihood [10]. However, the frequencies and amplitudes of variation required to achieve this effect under high speed milling conditions can be too demanding for the bandwidths of most spindle heads, leaving the helix as the most effective way to influence the period doubling lobes.

The effect of the helix on period doubling lobes was first described and experimentally validated by Zatarain et al. [11]. Using a simplified multifrequency model, they examined the effect of the helix on the amplitude of different harmonics of the directional factor leading to each of the flip lobes. They concluded that period doubling areas become closed instability islands because the helix reduces the amplitude of the harmonics to zero as the axial depth of cut approaches the helix pitch distance. This effect is illustrated in Fig. 1b. This finding was subsequently confirmed by semidiscretisation [12] and full discretisation [13] calculations. Subsequent studies experimentally verified this effect under different conditions [14] and in a hardware-in-the-loop environment [15]. However, none of these works was focused on the practical elimination of period doubling instability.

Therefore, the presence of period doubling lobes not only further limits the cutting capacity of the process but it also requires the use of more time-consuming stability prediction methods. Because of that, anticipating whether flip lobes are actually relevant for the potential range of spindle speed and axial depth of cut is crucial. This anticipation identifies if interrupted cutting is a danger at all for the considered range of machining conditions. Moreover, prior knowledge of the presence of period doubling lobes in the range of interest facilitates the selection of the most suitable stability prediction method. If no period doubling lobes are expected, ZOA should be the preferred option due to its almost instantaneous calculation. Conversely, if the presence of flip lobes is expected, discrete-time algorithms should be used for a precise calculation of the stability limits.

As reported by Munoa [16] and later by Khasawneh [17], as the helix pitch decreases, the flip lobes are not only divided into more and smaller islands, but they completely vanish below some critical pitch distance as shown in Fig. 1c. However, none of these studies has provided a closed-form formula for this limit. Since then, most of the recent studies have put the focus on designing special tool geometries to influence milling stability. Variable pitch straight fluted cutters generate a different family of PD lobes associated to the complete period of the tool [18], but do not directly eliminate them as in the case of helical tools. Other recent research works such as [19,20] have concentrated on irregular helix cutters, which have been proven effective to increase the stability of the cutting. For these helix geometries, whether tool runout is present [21] or not [22], the flip lobes also transform into a family of complete tool rotation period PD lobes. In [23], initial value time domain simulations were carried out to investigate the effect of irregular helix cutters presenting runout, revealing that not only complete tool rotation PD lobes were generated, but a complete family of period-n lobes. In any case, no investigation was done in these works on the necessary angle to remove period doubling lobes. Analytical expressions for irregular helix cutter design have been provided in [24]; however, the effect of flip lobes was omitted in the study and the focus was put on Hopf lobes.

In summary, there have been many research works studying the effect of the tool helix on the stability of milling. Numerous authors have confirmed that the helix angle of the cutter can shrink the period

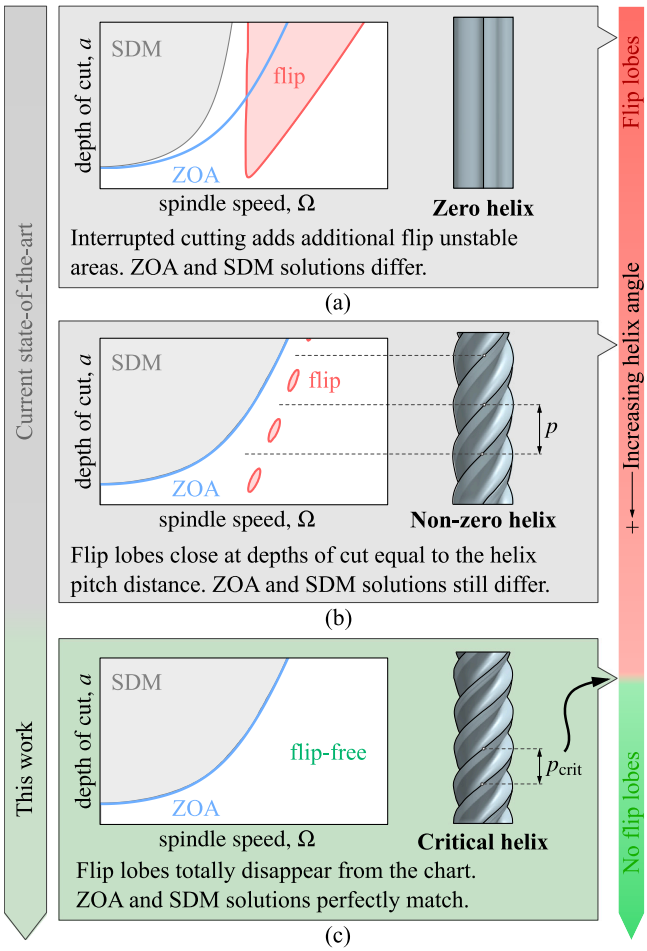

Fig. 1. Progress beyond the state-of-the-art. (a) Period doubling lobes add additional unstable areas to the already existing ones, making zeroth order prediction differ from semidiscretisation. (b) The helix transforms flip lobes into closed unstable islands separated by the helix pitch distance. (c) The main novelty of this work resides in the introduction of an analytical critical helix pitch distance below which no flip chatter can happen at all, making zeroth order algorithm exact.

doubling lobes as in Fig. 1b. Some have numerically anticipated that below a certain threshold helix pitch, the period doubling lobes are completely eliminated as in Fig. 1c. Recent research has concentrated on designing irregular helix cutters to simultaneously eliminate both Hopf and flip chatter by conducting extensive discrete-time calculations under various conditions. However, several research gaps remain unaddressed:

- What is the critical helix angle or pitch required to completely eliminate the likelihood of period doubling chatter? Is it possible to derive a closed-form formula for this limit?
- How do process and modal parameters influence this critical threshold?
- Is this critical threshold applicable to general multi degree of freedom cases?
- Do different orders of period doubling lobes require the same helix angle for elimination? Is it possible to eliminate all of them simultaneously?
- How can we determine if ZOA provides exact stability limits compared to SDM?

This paper aims to fill these research gaps by presenting a novel analytical formula to design milling cutter helices that completely eliminate period doubling chatter. By considering a single dominant mode multifrequency model, the expression relating the critical flip depth of cut of a helical tool to that of a tool with a straight helix is derived. This expression anticipates that if the helix pitch is below the critical flip depth of cut of the straight helix cutter multiplied by π, all the period doubling lobes will be eliminated. The formula provides clear guidance not only for avoiding flip chatter regardless of the machining parameters, but also to select the most efficient stability algorithm (ZOA or discrete-time) that can provide the stability boundaries with exactness. The expression and the existence of critical helix pitch values are then validated by semidiscretisation under three different milling scenarios representing potential real cases of high speed milling. Finally, the accuracy of the formula is also validated by means of experimental cutting tests.

2. Multifrequency model for single dominant mode with helical tool

So far, the flip stability properties of interrupted cutting have been studied in different manners. Some authors have studied the period doubling lobes by considering simplified intermittent turning processes with either time-domain [25] or frequency-domain [26] approaches. However, the intermittent turning process does not model the real time-varying chip thickness of real milling processes. In [27], an interrupted milling model with time-varying chip thickness was considered to achieve closed-form formulae for period doubling lobes, but with a straight fluted cutter.

To study the effect of the tool helix on the period doubling lobes and assess when the helix totally removes the period doubling lobes, in this study we present a planar (xy) milling model with a helical cutter and a single dominant flexible direction (see Fig. 2a).

2.1. Dynamic milling force for helical cutters

Consider a cylindrical milling cutter (see Fig. 2b) with a diameter D, Z regularly spaced teeth rotating at a constant spindle speed Ω (rpm), resulting in a tooth passing period $T_Z = 60/(\Omega Z)$ (s). Because of the tool rotation, the angular position of the first flute ($i = 1$) at the tool tip results in $\varphi_0(t) := \Omega\pi t/30$ (rad). The tool presents a constant and uniform helix pitch p distance that is related to the helix angle η as

$$p = \frac{D\pi}{Z\tan\eta}. \tag{1}$$

In flexible milling systems as the one considered here, the cutting generates stationary and dynamic (regenerative) cutting forces in the system. While the former is unavoidable and leads to static deflections and forced vibrations, it is the latter which gives lead to chatter instability depending on the process parameters and the system dynamics. After linearisation, the dynamic milling force projected onto the flexible direction q of the system results in

$$F(t, q(t), q(t-\tau)) = -a\,K_{tc}\,A(t)(q(t) - q(t-\tau)). \tag{2}$$

Variable τ represents the regenerative delay, which $\tau = T_Z$ for this simple case. Parameters a and K_{tc} stand for the axial depth of cut and tangential cutting coefficient, respectively.

Function $A(t)$ is the temporal directional factor, which represents the successive projection of the cutting force onto the flexible direction, and the projection of the flexible direction onto the chip thickness direction. In milling, the directional factor – and therefore the dynamic milling force – is time-dependent because the directions of the chip thickness and cutting forces constantly vary due to tool rotation and the cutting teeth entering and exiting the material.

Because a helical cutter has been considered here, the directional factor matrix must be calculated as the axial average directional factor matrix of every differential axial tool disks $B(t, z)$ within the axial depth of cut a, that is,

$$\begin{aligned} A(t, a) &= \frac{1}{a}\int_0^a B(t, z)\mathrm{d}z \\ &:= \frac{1}{a}\int_0^a \sum_{i=1}^{Z} g(\varphi_i(t, z))\, b(\varphi_i(t, z))\mathrm{d}z. \end{aligned} \tag{3}$$

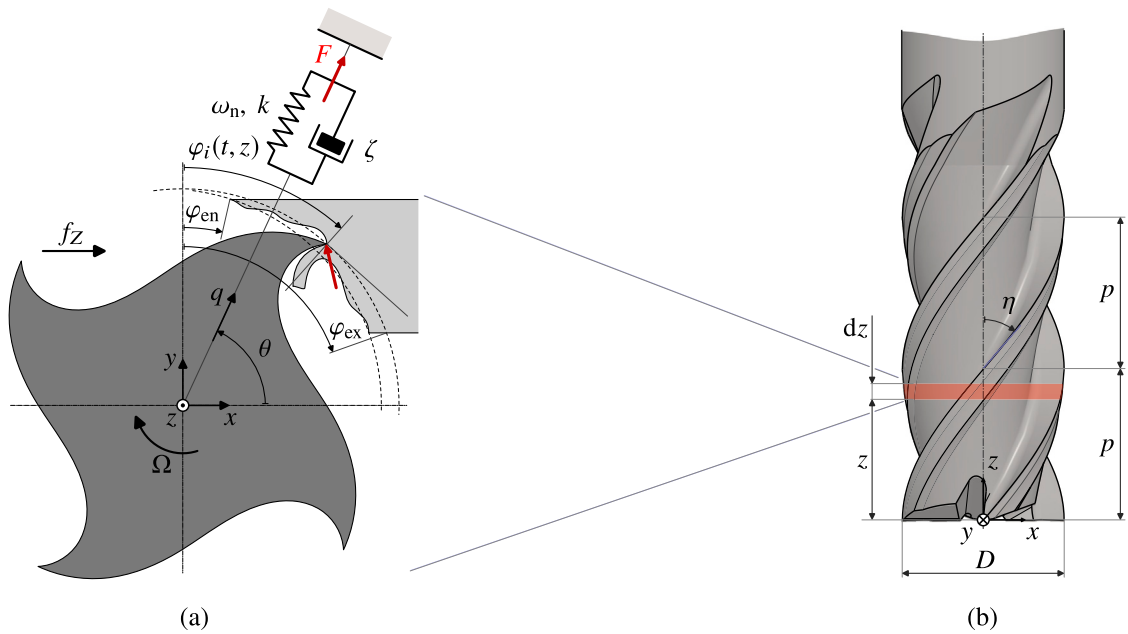

Fig. 2. Sketch of the regenerative milling model with helical tool consideration. (a) Planar (xy) regenerative milling taking place in a differential axial slice of height dz at a height from the tool tip z. (b) Representation of a regular helical tool. Here the relation between the helix angle and the helix pitch distance can be noticed: the helix pitch distance is defined as the axial distance between two consecutive teeth measured along the generatrix of the tool cylinder.

The angular position distribution $\varphi_i(t,z)$ (rad), which represents the angular position of the ith flute at the instant t and axial height z, ultimately reflects the smoothing effect of helical flutes on the fluctuations of the directional factor and, consequently, on period doubling stability. Here, due to the regular helix, for a tool rotation angle $\varphi_0(t)$ it can be calculated as

$$\varphi_i(t,z) = \varphi_0(t) + (i-1)\varphi_p - \varphi_\eta(z), \tag{4}$$

where $\varphi_p := 2\pi/Z$ is the pitch angle (rad), while

$$\varphi_\eta(z) := \frac{2\tan\eta}{D} z = \frac{2\pi}{Z\,p} z \tag{5}$$

is the lag angle (rad).

Consequently, as evident in (4) and (5), if the number of teeth Z remains constant, it is the helix pitch distance p that encapsulates the smoothing effect of the helix. Thus, although the helix angle η is the industrially adopted parameter for describing the helix of an end-mill, the tool helix pitch distance must be studied as the smoothing control parameter, which can be altered by either tuning the diameter or by selecting different helix angles.

Then, the contribution of each tooth to the directional factor is determined by:

$$b(\varphi) = \sin(\theta+\varphi)\left(k_r \sin(\theta+\varphi) + \cos(\theta+\varphi)\right), \tag{6}$$

where $k_r := K_{rc}/K_{tc}$ represents the cutting force ratio. g is the screen function

$$g(\varphi) = \begin{cases} 1, & \text{if} \quad \varphi_{en} < \varphi \bmod 2\pi < \varphi_{ex}, \\ 0, & \text{otherwise}, \end{cases} \tag{7}$$

which determines whether the cutting edge is in or out of the radial engagement limits $[\varphi_{en}, \varphi_{ex}]$.

The resulting directional factor is time-periodic $A(t) = A(t+T_Z)$ with period equal to the tooth passing period. As seen in (2), it directly modulates the amplitude of the dynamic milling force giving lead to chatter instability. While its mean value it is related to Hopf instability, it is its time-periodic variations which give lead to period doubling instabilities. Therefore, the aim of the next section will be to study the effect of the helix on the variations of the directional factor.

2.2. Effect of the helix on the dynamic milling force

In the previous section it has been noted that the variations of the dynamic milling force due to the variations of $A(t)$ can produce period doubling chatter. Because of its time-periodicity, these variations of the dynamic milling force can be studied in the frequency domain by means of the Fourier expansion of the infinite harmonics of $A(t)$ at the fundamental frequency $\omega_Z = 2\pi/T_Z$ (in rad/s):

$$A(t) = \sum_{r=-\infty}^{\infty} A_r\, \mathrm{e}^{\mathrm{i}r\omega_Z t}. \tag{8}$$

Each of the r harmonics of the directional factor are calculated as

$$\begin{aligned} A_r(a) &= \frac{1}{T_Z}\int_0^{T_Z} A(t,a)\,\mathrm{e}^{-\mathrm{i}r\omega_Z t}\,\mathrm{d}t \\ &= \frac{1}{T_Z}\int_0^{T_Z} \frac{1}{a}\int_0^a B(t,z)\,\mathrm{d}z\,\mathrm{e}^{-\mathrm{i}r\omega_Z t}\,\mathrm{d}t. \end{aligned} \tag{9}$$

Changing the integral limits to a whole tool period:

$$A_r(a) = \frac{1}{Z\,T_Z}\int_0^{Z\,T_Z} \frac{1}{a}\int_0^a B(t,z)\,\mathrm{d}z\,\mathrm{e}^{-\mathrm{i}r\omega_Z t}\,\mathrm{d}t. \tag{10}$$

Then, by expressing (10) in terms of angular position and flipping the order of integration:

$$A_r(a) = \frac{1}{2\pi}\frac{1}{a}\int_0^a \int_{(i-1)\varphi_p}^{2\pi+(i-1)\varphi_p} \sum_{i=1}^{Z} g(\varphi_i) b(\varphi_i) \mathrm{e}^{-\mathrm{i}rZ\varphi_i}\,\mathrm{d}\varphi\,\mathrm{d}z. \tag{11}$$

By the definition of the angular position of each flute i and axial disk at z in (4), the term $\mathrm{e}^{-\mathrm{i}rZ\varphi_i}$ can be split into three terms: the first one corresponding to the tool rotation, the second one corresponding to the flute pitch and the last one corresponding to the lag angle:

$$\mathrm{e}^{-\mathrm{i}rZ\varphi_i} = \mathrm{e}^{-\mathrm{i}rZ\varphi}\,\mathrm{e}^{-\mathrm{i}rZ(i-1)\varphi_p}\,\mathrm{e}^{\mathrm{i}rZ\varphi_\eta(z)}. \tag{12}$$

Then, as $\mathrm{e}^{-\mathrm{i}rZ(i-1)\varphi_p} = \mathrm{e}^{-\mathrm{i}r(i-1)2\pi} = 1$ and by the definition of the lag angle in (5),

$$\mathrm{e}^{-\mathrm{i}rZ\varphi_i} = \mathrm{e}^{-\mathrm{i}rZ\varphi}\,\mathrm{e}^{\mathrm{i}2\pi r\frac{z}{p}}. \tag{13}$$

Considering that out of the radial engagement limits $[\varphi_{en}, \varphi_{ex}]$ the value of the screen function is zero, (11) can be expressed as

$$A_r(a) = \frac{Z}{2\pi}\frac{1}{a}\int_0^a \int_{\varphi_{en}}^{\varphi_{ex}} b(\varphi)\mathrm{e}^{-\mathrm{i}rZ\varphi}\mathrm{d}\varphi\,\mathrm{e}^{\mathrm{i}2\pi r\frac{z}{p}}\mathrm{d}z. \tag{14}$$

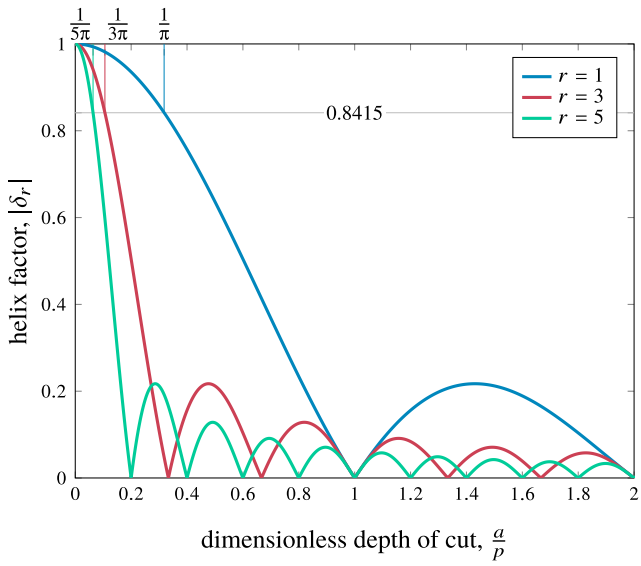

Fig. 3. Absolute value of the helix factor. The helix factor relates the amplitude of the directional factor of a straight-fluted cutter to that of a helical tool. As its amplitude drops to zero when the a/p ratios matches multiples of $1/r$, the flip lobes become closed islands at depths of cut equal to the helix pitch divided by r.

Finally, in (14) two different terms can be devised:

$$A_r(a) = \frac{Z}{2\pi} \underbrace{\int_{\varphi_{\mathrm{en}}}^{\varphi_{\mathrm{ex}}} b(\varphi) \mathrm{e}^{-\mathrm{i}rZ\varphi} \mathrm{d}\varphi}_{=:\beta_r} \underbrace{\frac{1}{a} \int_0^a \mathrm{e}^{\mathrm{i}2\pi r \frac{z}{p}} \mathrm{d}z}_{=:\delta_r(a)}, \tag{15}$$

namely β_r, which is the rth straight helix harmonic of the directional factor and

$$\delta_r(a) = \frac{1 - \mathrm{e}^{-\mathrm{i}2\pi r \frac{a}{p}}}{\mathrm{i}2\pi r \frac{a}{p}}, \tag{16}$$

the rth helix factor [11]. The complex-valued helix factor δ_r relates the harmonics of the directional factor of a straight fluted cutter to the ones corresponding to a cutter with regular helix. The evolution of the amplitude of the helix factor for the different harmonics r and increasing ratio between the axial depth of cut and the helix pitch a/p are shown in Fig. 3.

Therefore, depending on the dimensionless a/p ratio the helix factor will attenuate each harmonic of the dynamic milling force as

$$F(t) = \sum_{r=-\infty}^{\infty} F_{r,\eta} \, \mathrm{e}^{\mathrm{i}r\omega_Z t} = \sum_{r=-\infty}^{\infty} \delta_r F_r \, \mathrm{e}^{\mathrm{i}r\omega_Z t}, \tag{17}$$

where $F_{r,\eta}$ are the harmonics of the dynamic milling force with a helical tool and F_r the harmonics of the equivalent straight fluted tool.

2.3. Closed form solution of flip lobes for straight helix

Prior to evaluating the effect of the helix on the flip lobes, it is important to study the stability properties of straight fluted cutters in the frequency domain. As $\delta_r = 1$ for straight helix, the directional factor no longer depends on the axial depth of cut and can be related to the straight helix directional factor as, i.e., $A_r = \frac{Z}{2\pi}\beta_r$.

To do so, the critical case when the system becomes unstable at a chatter frequency ω_{c} (in rad/s) plus its infinite r modulations at the tooth passing frequency ω_Z is considered:

$$q(t) = \sum_{r=-\infty}^{\infty} q_r \, \mathrm{e}^{\mathrm{i}(\omega_{\mathrm{c}} + r\omega_Z)t}. \tag{18}$$

Thus, the regenerative term becomes:

$$q(t) - q(t-\tau) = \left(1 - \mathrm{e}^{-\mathrm{i}\,\omega_{\mathrm{c}}\,\tau}\right) \sum_{r=-\infty}^{\infty} q_r \, \mathrm{e}^{\mathrm{i}(\omega_{\mathrm{c}} + r\,\omega_Z)t}. \tag{19}$$

To relate each of the r modulations of the regenerative force of the straight helix cutter F_r and the dynamic displacement q_r, the receptance function evaluated at each modulation frequency $\omega_{\mathrm{c}} + r\omega_Z$ is used, that is, $q_r = \Phi(\omega_{\mathrm{c}} + r\omega_Z)F_r = \Phi_r F_r$. For this unimodal case with natural frequency ω_{n}, reflected modal stiffness k and damping ratio ζ, the receptance can be synthetised as

$$\Phi(\omega) = \frac{1}{k} \frac{\omega_{\mathrm{n}}^2}{\omega_{\mathrm{n}}^2 - \omega^2 + 2\mathrm{i}\omega\omega_{\mathrm{n}}\zeta}. \tag{20}$$

With all this, the characteristic equation for the rth harmonic can be formulated as

$$q_r = -a\,K_{tc}\left(1 - \mathrm{e}^{-\mathrm{i}\omega_{\mathrm{c}}\tau}\right) \sum_{l=-\infty}^{\infty} \Phi_r \, B_{r-l} \, q_l. \tag{21}$$

To ensure the stability of the system all the harmonics must be considered. This leads to an infinite dimensional Hill-kind eigenvalue problem that represents both Hopf and period doubling instabilities. One of the possible approaches to obtain the parametric root-crossing curves from (21) is the multifrequency [5,28] method. This requires setting a fixed set of harmonics and performing a nested ω_{c}, ω_Z scan to obtain the real depths of cut for each combination.

Nevertheless, period doubling lobes can be analytically obtained from (21) without the need for a classic numerical multifrequency calculation. For systems with a single dominant mode, period doubling lobes are generated when the system is being shaken near its resonance by both the mean term and a rth modulation, positioned symmetrically about the ordinate axis in the frequency response diagram [3]. This gives rise to an analytical dominant chatter frequency distribution $\omega_{\mathrm{c}} = \frac{r\omega_Z}{2}$ during period doubling chatter. When this is the case, the regenerative term becomes

$$1 - \mathrm{e}^{-\mathrm{i}\omega_{\mathrm{c}}\tau} = 1 - \mathrm{e}^{-\mathrm{i}\omega_{\mathrm{c}}\frac{2\pi}{\omega_Z}} = \begin{cases} 0, & \text{if} \quad r = 0, 2, 4, \ldots, \\ 2, & \text{otherwise}, \end{cases} \tag{22}$$

In other words, each of the flip lobes is dominantly generated by one of the odd $(-1, -3, -5, \ldots)$ modulations of the regenerative force and the 0th (mean) harmonic.

Accordingly, as in [27], each flip lobe can be approximated by only considering the 0th and the corresponding negative rth harmonic in (21), which reduces the infinite-dimensional eigenvalue problem to

$$\begin{bmatrix} q_{-r} \\ q_0 \end{bmatrix} = \Lambda \begin{bmatrix} \Phi_{-r} & 0 \\ 0 & \Phi_0 \end{bmatrix} \begin{bmatrix} \beta_0 & \beta_{-r} \\ \beta_r & \beta_0 \end{bmatrix} \begin{bmatrix} q_{-r} \\ q_0 \end{bmatrix}, \tag{23}$$

where $\beta_{-r} = \bar{\beta}_r$ denotes the complex rth straight fluted harmonic of the directional factor. Then, critical depth of cut for period doubling chatter can be approximated as

$$a_{r,\min} \cong \frac{2\pi k\zeta\,\lambda_{r,\min}}{K_{tc}\,Z\,|\beta_r|}, \tag{24}$$

Here $r_{\beta,r} := |\beta_r|/|\beta_0|$ represents the amplitude ratio between the rth straight fluted harmonic of the directional factor and its mean value. $\lambda_{r,\min}$ represents the dimensionless flip frequency at the minimum

$$\lambda_{r,\min} := \frac{\omega_{\mathrm{c,min}}}{\omega_{\mathrm{n}}} \cong \begin{cases} \sqrt{1 + \left(\frac{\zeta}{r_{\beta_r}}\right)^2} + \frac{\zeta}{r_{\beta_r}}, & \beta_0 > 0, \\ \sqrt{1 + \left(\frac{\zeta}{r_{\beta_r}}\right)^2} - \frac{\zeta}{r_{\beta_r}}, & \beta_0 < 0. \end{cases} \tag{25}$$

For more details on the derivation of these expressions, refer to [27]. In interrupted milling cases, where large $r_{\beta,r}$ ratio and low damping level is usually encountered, $\lambda_{r,\min} \approx 1$, therefore,

$$a_{r,\min} \approx \frac{2\pi k\zeta}{K_{tc}\,Z\,|\beta_r|}. \tag{26}$$

For single mode systems, the expression in (24) provides very accurate results for the first ($r = 1$) flip lobe, as when sampling chatter frequencies near the natural frequency (approximately when the minimum is being produced) the truncated harmonics in (21) sample the receptance at higher frequencies where the dynamic flexibility is decaying. As higher flip order is considered, a wider set of harmonics sample the receptance at frequencies between zero and ω_n, leading to a larger deviation of (24).

2.4. Optimal selection of helix pitch for flip lobe suppression

Eq. (26) showed that the minimum of the rth flip lobe of a straight fluted cutter is inversely proportional to the absolute value of the corresponding harmonic rth. In the case of helical cutters, the directional factor is attenuated by the helix factor, whose value, in turn, depends on the actual depth of cut, that is, $a_{r,\min,\eta}$. This leads to the following implicit limit:

$$a_{r,\min,\eta} \cong \frac{2\pi k\zeta}{K_{tc} Z|\beta_r||\delta_r(a_{r,\min,\eta})|} = a_{r,\min}\frac{1}{|\delta_r(a_{r,\min,\eta})|}, \tag{27}$$

The absolute value of the helix factor can be then computed as

$$|\delta_r| = \frac{\sin \pi r \frac{a}{p}}{\pi r \frac{a}{p}}, \quad \forall \frac{a}{p} < 1. \tag{28}$$

Fig. 3 shows the evolution of the absolute value of the helix factor for different odd harmonics — the ones generating flip lobes. As reported by Zatarain et al. [11], for the rth flip lobe, when the axial depth of cut matches p/r the directional factor is completely canceled and therefore the flip lobe or island closes at that depth of cut. This leads to smaller and smaller islands as higher flip order is considered.

Intuitively, one could think that if $p = ra_{r,\min}$, the corresponding r flip lobe completely vanishes. Indeed, when this happens, $|\delta_r(a_{r,\min})| = |\delta_r(\frac{p}{r})| = 0$ and thus the harmonic r of regenerative milling force giving rise to the corresponding flip lobe is already canceled when the axial depth of cut reaches the flip minimum. Consequently, at the critical depth of cut there is no variation of the regenerative milling force at all and for greater depths of cut the amplitude of the regenerative milling force is not sufficient to lead to flip instability.

However, (29) and (28) anticipate that to cancel the period doubling instability at the minimum depth of cut of the lobe r it is not necessary to totally cancel the corresponding harmonic. By combining (29) and (28), the following closed form formula of the critical depth of cut of helical tool as a function of the flip critical depth of cut of the straight fluted cutter:

$$a_{r,\min,\eta} = \frac{p}{\pi r}\arcsin\left(\pi r \frac{a_{r,\min}}{p}\right). \tag{29}$$

From (29) it is possible to infer that there exists a critical helix pitch for which $a_{r,\min,\eta}$ is no longer real and therefore no minimum or flip lobe exists below this helix pitch value:

$$p_{r,\text{crit}} = \pi r a_{r,\min}. \tag{30}$$

Therefore, (30) indicates that to totally cancel the corresponding r flip lobe, it is sufficient to achieve a decrease of the amplitude corresponding harmonic down to $|\delta_r(\frac{p}{\pi r})| = \sin 1 = 0.8415$ at the flip critical, as shown in Fig. 3.

For most of the cases, the first period doubling lobe leads to the lowest minimum. Additionally, as higher flip lobe order is considered, the lobe vanishes for higher p values. Consequently, it is possible to state that if $p < \pi a_{1,\min}$, no period doubling lobes can exist in the whole spindle speed range. Alternatively, if the diameter is fixed, a critical helix angle can be defined as

$$\eta_{r,\text{crit}} = \arctan\frac{D}{Zra_{r,\min}}. \tag{31}$$

Analytical formulae (30) and (31) are highly significant for two reasons: Firstly, by only knowing the minimum flip depth of cut of an straight helix cutter $a_{1,\min}$, the minimum helix value ensuring that no chatter flip will occur regardless of the spindle speed or axial depth of cut can be calculated. This $a_{1,\min}$ can be determined experimentally or from (24) by knowing modal parameters of the dominant mode and the conditions of radial engagement (resulting in β_0, β_1 and r_{β_1}). Moreover, as long as $a_{1,\min}$ is accurately calculated, arbitrary xyz flexible milling cases can be considered with any kind of cutting force characteristics. On the other hand, it is also valuable when selecting the algorithm to predict process stability. If the helix pitch is below the critical value given in (30) for the first lobe, there will be no flip effect, allowing the ZOA method to be used with precise accuracy and very low computational cost. If the step value is higher, flip islands will exist, necessitating time-domain [7] or multifrequency [11] methods to obtain the stability diagram with exactness.

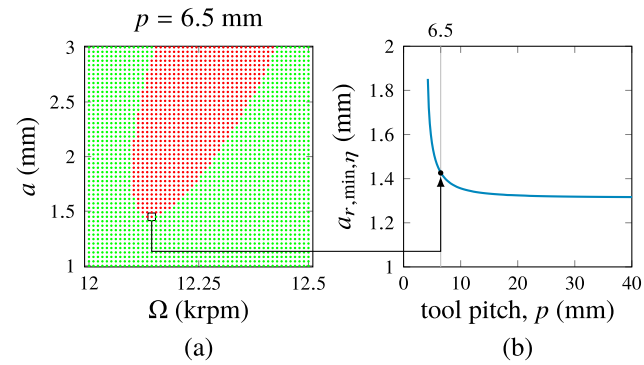

Fig. 4. Procedure to calculate $a_{r,\min,\eta}$ by means of semidiscretisation. (a) Grid of stable and unstable points close to the minimum of the corresponding lobe for a fixed value of the pitch. (b) Obtained curve by extracting the minima from a vector of values of p.

3. Numerical validation

To validate the accuracy and effectiveness of the formulae presented in this paper, in this section the results of the formula are validated against the results of semidiscretisation calculations. Semidiscretisation algorithm is used in two ways to validate the introduced findings. On the one hand, traditional Ω-a stability lobe diagrams are obtained for discrete values of the helix pitch/angle. The unstable regions associated to period doubling bifurcations are then detected by checking the Floquet multipliers. If pure real negative multipliers are out of the unit circle in the complex plane, flip unstability takes place [7]. The order of the flip lobe is then determined by checking the chatter frequency [29] as $r = \omega_c/\omega_Z$.

On the other hand, to validate the formula (29) relating the straight helix minimum to that of a helical cutter, for a discrete vector of values p (see Fig. 4b), a fine grid (100×100) of Ω-a points is generated in the vicinity of the theoretical minimum of the corresponding flip lobe order (see Fig. 4a). The minimum depth of cut associated to that value of p is then obtained as the minimum unstable flip depth of cut in that fine mesh.

To validate the applicability of the formula introduced in this paper, the formula is tested on three different high speed milling scenarios. Firstly, the formula is tested on a single degree of freedom taking place in a real thin wall milling case. Next, two different two degree of freedom cases are studied. The first one represents a slender milling tool case where the same high frequency mode exists in perpendicular directions. Finally, a machine tool structure chatter case is introduced, which presents two different low frequency modes at perpendicular directions.

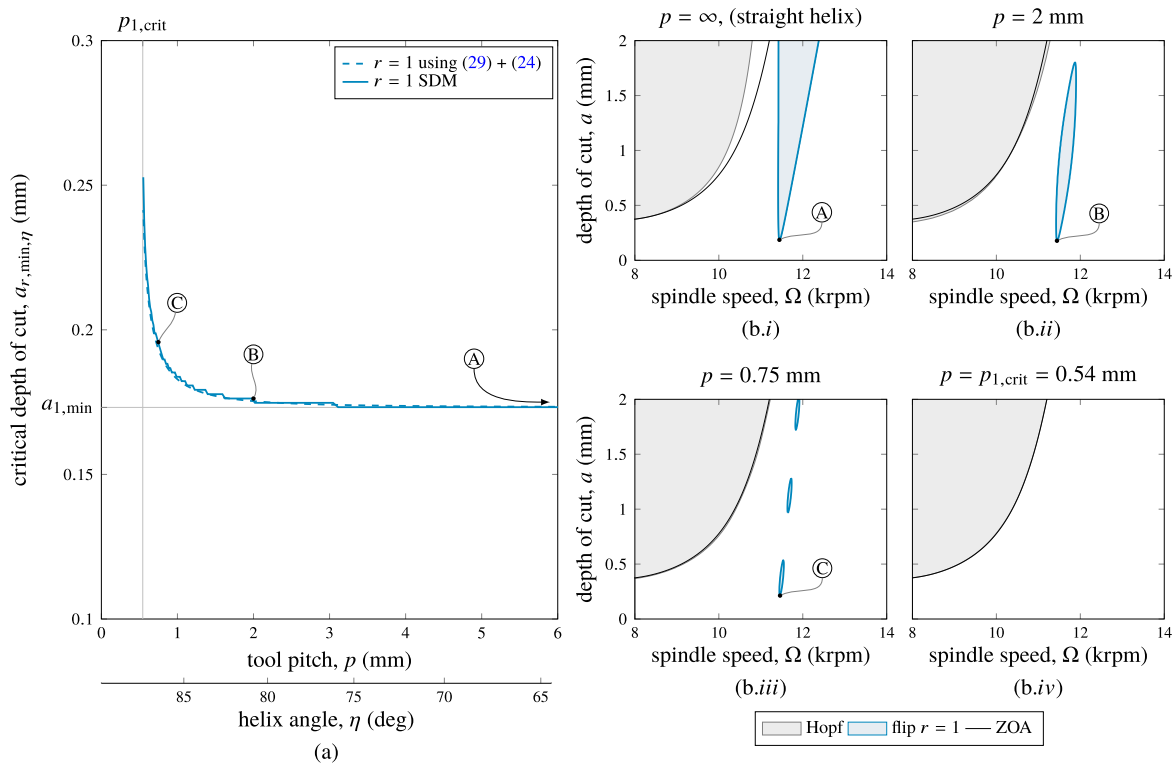

Fig. 5. Results of the numerical validation of the formula for the first flip lobe for a thin wall milling case (one degree of freedom). (a) Variation of the critical flip depth of cut, calculated by the analytical formulae (29) (using the analytical critical $a_{1,\min}$ from (24)) and semidiscretisation, showing almost equal results. (b) Stability lobe diagrams obtained by semidiscretisation for different values of the helix pitch. As pitch decreases, the flip lobe is divided into islands separated by the pitch distance. When the critical in (30) is reached, the lobe is completely suppressed.

3.1. One degree of freedom case: thin wall peripheral milling

Peripheral milling of thin-walled parts represents the most evident case of single degree of freedom cases. Thin-walled features are typically milled by performing successive passes perpendicular to their 'thickness direction' (y direction, $\theta = 90$ deg), which is the direction of maximum flexibility. During semi-finishing or finishing operations under high speed conditions, period doubling chatter can limit the stable range of spindle speeds, especially if the tooth passing frequency is above any of the natural frequencies of the part.

To validate the applicability of the findings for this scenario, the peripheral thin wall milling real case from [30] is considered. The thin-walled steel part presents a dominant bending mode along y direction, that is, perpendicular to the feed direction. The mode presents a natural frequency $\omega_n = 380$ Hz, relative damping $\zeta = 0.4\%$ and modal stiffness $k = 3.02$ N/μm. The process is a finishing pass with 1.25% radial engagement under up-milling strategy carried out with a $D = 16$ mm cutter with $Z = 4$ flutes. The cutting coefficients of the linear cutting force model are $K_{tc} = 1795$ MPa and $K_{rc} = 721$ MPa.

First flip lobe

First, the focus is made on the first ($r = 1$) period doubling lobe, for which the developed formulae are expected to be exact. On the one hand, Fig. 5a shows the variation of the minimum flip depth of cut of the first flip lobe for decreasing value of the helix pitch distance p. For a constant diameter and number of teeth, this decrease can be achieved by increasing the helix angle as shown in the helix angle scale below. The values obtained with the application of analytical formulae (24) and (29) (- -) are compared to the ones obtained by SDM (——). On the other hand, Fig. 5b shows the stability limits in the spindle speed region dominated by the $r = 1$ flip lobe for different values of the helix pitch, obtained by the ZOA solution (——) and the exact SDM solution (▭). The SDM stability boundaries of the period doubling lobes are highlighted in (▭). For the straight fluted cutter, an evident deviation between the ZOA and SDM boundaries can be noticed because of the flip lobe and mode interaction effects [3].

The results show that the combination of formulae (24) and (29) provides very close prediction of the flip minimum for all the values of the helix pitch. The minimum depth of cut for straight fluted cutter $a_{1,\min}$ is predicted by (24) with −0.1% error, while the critical pitch $p_{1,\mathrm{crit}}$ is predicted with a −0.2% error.

When the helix pitch is far from the critical value $p_{1,\mathrm{crit}}$ provided by (30), the flip lobe closes to instability islands at depths of cut equal to the tool pitch (see Fig. 5b.ii) as in [11], but its critical depth of cut suffers negligible variation. As the helix pitch approaches the critical value $p_{1,\mathrm{crit}}$ provided by (30), the critical depth of cut suddenly increases until the flip lobe totally vanishes when the critical pitch is reached, as shown in Fig. 5b.iv. Therefore, if $p = p_{1,\mathrm{crit}} = \pi a_{1,\mathrm{crit}}$, the ZOA and SDM solutions perfectly match.

It is important to remark that while (29) predicts an increase of the critical depth of cut, by the time the formula predicts noticeable increase of the critical depth of cut with respect the straight helix case, the flip instability island is already reduced to a very small area as seen in Fig. 5b.iii. Therefore, the most relevant outcome of formula (29) is not the prediction of the critical as it is, but the prediction of the limit when the flip lobe will totally disappear.

Therefore, for this high speed thin wall milling scenario a very reduced value of the critical pitch is required (0.54 mm) for total elimination of period doubling chatter, for which the considered cutter diameter would suppose a practically unfeasible helix angle of 87.5°. Indeed, for very flexible or very lowly damped systems such as thin-walled parts, depending on the cutting conditions, the flip minimum of the straight helix cutter can be very low and hence a very large helix angle could be necessary to completely suppress the lobes. In such cases, where a commercial helix angle cannot be chosen to completely

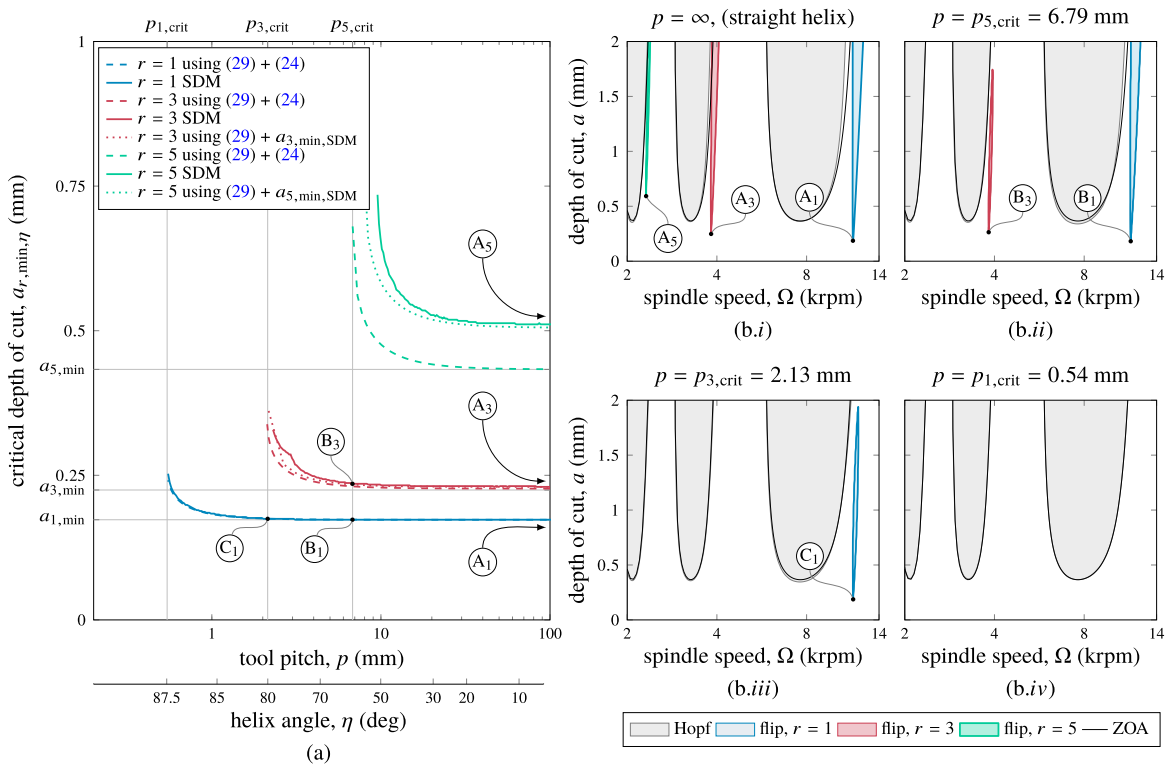

Fig. 6. Results of the numerical validation of the formula for the higher order flip lobes for a thin wall milling case. (a) Variation of the critical flip depths of cut. When the analytical critical $a_{r,\mathrm{min}}$ from (24) is used, significant deviation is found. This deviation is much reduced when the minima from semidiscretisation $a_{r,\mathrm{min,SDM}}$ are used. (b) Stability lobe diagrams for different values of the helix pitch. As the helix pitch decreases, the different flip lobes vanish. When the critical helix pitch of the $r = 1$ lobe is reached, all the flip lobes, including the higher order ones, are eliminated.

eliminate the lobe, a compromise between the selection of a commercially available helix angle and a decrease of the radial immersion could be achieved by using the analytical formulae (24) and (31).

Higher flip lobe orders

It has been observed that for a one degree of freedom system, although it results in an impractical helix angle, the prediction over the $r = 1$ is almost exact in both critical depth of cut and the critical pitch predictions. Even if the first flip lobe is the one leading to the widest unstable regions, if the spindle speed is lower, higher order flip lobes can be encountered. To study the validity of the theoretical prediction of the critical depth of cut and critical pitch for higher flip orders, the same thin wall milling case is considered but studying the evolution of the first three flip lobes, namely $r = 1$, $r = 3$ and $r = 5$.

Fig. 6a shows the results of the comparison of semidiscretisation curves for the three lobes (——, for $r = 1$, —— for $r = 3$ and —— for $r = 5$) compared to the analytical ones provided by the analytical formulae (24) and (29) (- - , - - and - -). The charts are provided in logarithmic x scale to compare the three lobes. In the charts of Fig. 6b the stability lobe diagrams obtained by SDM and ZOA are depicted for a straight fluted cutter and the three critical pitch values calculated according to (30).

The results show that for higher flip lobes there exists larger deviations of the analytical prediction with respect to the SDM calculations. For the $r = 3$ lobe, the prediction of the straight fluted minimum $a_{3,\mathrm{min}}$ by (24) already in a −1.3% deviation with respect to SDM. Based on that critical depth of cut, the prediction of the critical pitch $p_{3,\mathrm{crit}}$ given by (30) makes an error of −6.1% (the lobe would have disappeared for lower helix). For the $r = 5$ lobe, the minimum of the straight flute $a_{5,\mathrm{min}}$ produces a −14.3% deviation, while the error prediction of the critical pitch $p_{5,\mathrm{crit}}$ rises up to −28.1% (the lobe would have disappeared for lower helix). The stability diagrams for each of the criticals in Fig. 6b corroborate these results, as in Fig. 6b.ii the $r = 5$ lobe is not present and in Fig. 6b.iii the $r = 3$ lobe has vanished.

The cause of these deviations is the effect of the intermediate harmonics already mentioned at the end of Section 2.3: as higher flip lobe order r is considered, a wider set of harmonics sample the receptance between zero and ω_{n}. This generates increasing deviations in the prediction of the critical flip depth of cut of a straight fluted cutter (24), which is later used in (29) to calculate the minimum with a helical cutter.

Consequently, rather than using the analytical calculation in (24) to calculate the minimum of the straight helix case $a_{r,\mathrm{min}}$, one could think of using the straight fluted criticals numerically obtained by semidiscretisation $a_{r,\mathrm{min,SDM}}$ in (29) to study their variation when helix is introduced. The resulting criticals of this SDM + (29) calculation are depicted in Fig. 6a (····· and ····). By doing this calculation, the error of the prediction of the critical pitch for the three lobes is reduced to −4.9% and −16.1%, respectively.

In terms of the practical selection of the cutting tool, the analytical formula in (31) predicts that a minimum helix angle of 80.3° is required to guarantee no period doubling chatter at spindle speeds below the first flip lobe. To remove the $r = 5$ and higher flip lobes, a helix angle of 61.6° would be required. Accordingly, for cases where a very low critical period doubling depth of cut takes place due to the high dynamic flexibility, the presence of higher order period doubling lobes can be guaranteed even for commercial helix values.

Nevertheless, it is important to remember that beyond the accuracy of (29), the validity of the most important finding of the paper still applies even with the consideration of higher order flip lobes: when the value of the critical pitch of the first lobe $p_{1,\mathrm{crit}}$ is reached (which can be accurately predicted), all the flip lobes vanish and therefore the ZOA and SDM solutions perfectly match for entire the spindle speed range, as shown in Fig. 6b.iv. This implies that with a simple calculation of the critical depth of cut of straight helix cutter in (24) and the cutter

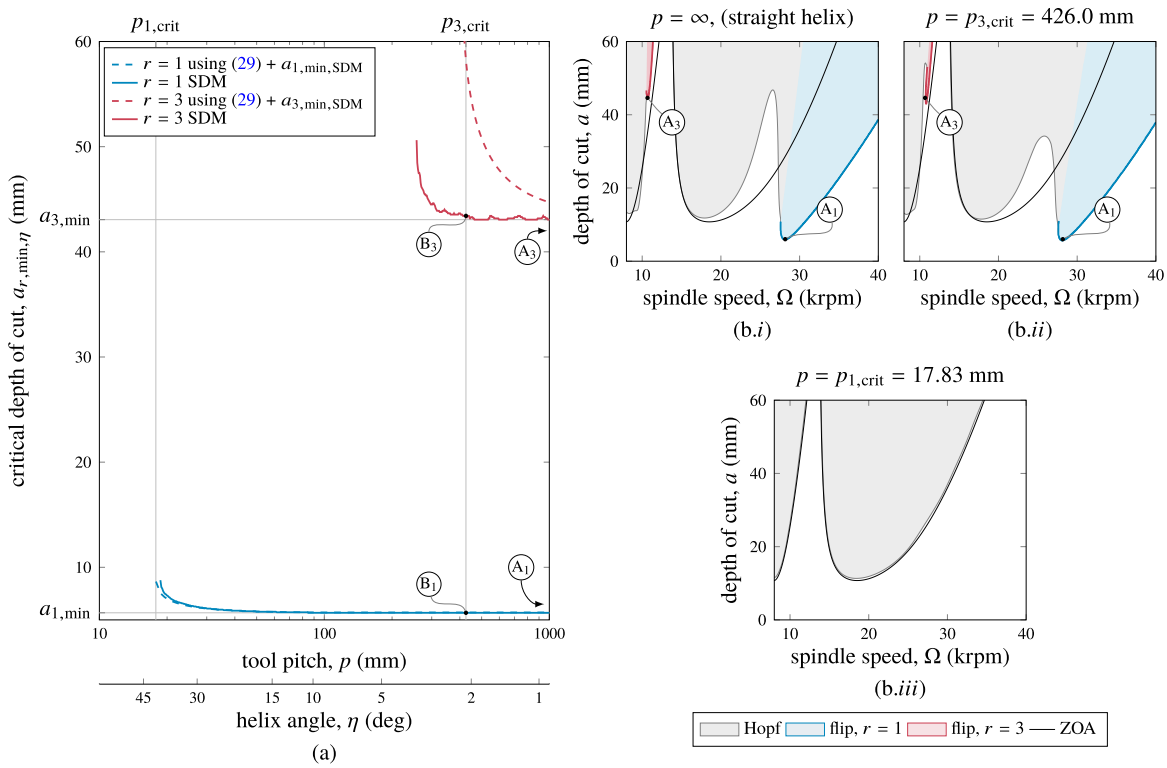

Fig. 7. Results of the numerical validation of the formula for the slender tool chatter case (two degree of freedom). (a) Variation of the critical flip depths of cut, calculated by the analytical formulae (29) (using the minima from semidiscretisation $a_{r,\mathrm{min,SDM}}$) and semidiscretisation. (b) Stability lobe diagrams for different values of the helix pitch. Once the critical pitch of the first lobe is reached, all the flip lobes are removed.

helix pitch, it would be possible to select the most suitable stability algorithm to construct the chart accurately.

3.2. Two degree of freedom cases

In the previous section the validity of the findings over a thin wall milling case was analyzed, which represents a single degree of freedom case. For such a case, the formulae produced almost exact results for the first lobe, while it produced deviations for higher lobes. Nonetheless, in the industry cases with more than one flexible cartesian directions and multiple are often encountered. As in one degree of freedom cases, period doubling chatter can limit the productivity when performing high speed milling operations under interrupted cutting conditions.

For multiple degree of freedom systems, analytical formulae (29) and (30) can be still used. The main issue resides in the calculation of the critical flip depth of the equivalent straight fluted cutter $a_{r,\mathrm{min}}$. Analytical formula (24) used for single degree of freedom and single mode cases can be used for multiple mode cases as far as all the relevant modes present the same flexible direction, by applying it mode-per-mode. However, when more than one cartesian flexible directions are present, the scalar directional factor harmonics β_r turn into matrices $\boldsymbol{\beta}_r$, which prevent the derivation of an analytical formula for $a_{r,\mathrm{min}}$ from the eigenvalue problem in (23).

Consequently, for multiple degree of freedom cases the application of (29) and (30) must be carried out with the numerical minimum from semidiscretisation $a_{r,\mathrm{min,SDM}}$. Alternatively, this minimum could be calculated by other frequency domain methods [5,6]. To test this, in this section two relevant industrial milling cases are considered. First, a slender tool case is studied, where the same high frequency mode exists along two perpendicular directions. Then, a machine tool structure chatter case is studied, where two different modes exist in perpendicular directions.

Slender tool chatter case

Regenerative chatter cases originated by the use of slender cutting tools is one of the typical cases in milling, where high frequency tool bending modes with similar modal parameters but in perpendicular directions in the xy plane usually arise. When milling at very high spindle speeds (>20,000 rpm) such as in high speed aluminum machining, the tooth passing frequencies can be above the natural frequencies of the tool and therefore lead to period doubling chatter.

To validate the helix selection formulae for this milling scenario, the slender tool case from [31] is taken as a reference. In this case, a 20 mm 4 teeth slender end mill produces two ω_n = 898.3 Hz, ζ = 4.02% and k = 6.41 N/μm modes in perpendicular directions. The considered process is an aluminum finishing or semi-finishing down-milling operation with 5% radial immersion. The considered linear cutting coefficients are K_{tc} = 800 MPa and K_{rc} = 240 MPa.

Fig. 7a shows the evolution of the critical depths of cut of lobes r = 1 and r = 3 obtained by (29) and $a_{r,\mathrm{min,SDM}}$ (- - , - -) compared to the ones directly obtained by semidiscretisation (——, ——). The results show that the critical pitch to completely eliminate flip chatter is predicted with an error of −4.5%. Consequently, when this helix is selected, all the flip lobes are removed as in Fig. 7c.iii and ZOA and SDM solutions perfectly match. On the contrary, for the r = 3 lobe, the critical is predicted with an error of 65.4% and therefore this lobe is not removed when this pitch is selected as in Fig. 7c.ii.

Therefore, the presented formula is still effective for this special two degree of freedom case. Even though, as in the previous case, the prediction for higher flip lobes is not exact, the predicted pitch to remove all the flip lobes results in a minimum helix angle of 41.4°, which can be reached by commercially available tools. Nonetheless, as previously mentioned, for multiple degree of freedom systems the calculation of the straight fluted minimum $a_{r,\mathrm{min}}$ has to be carried out numerically. Still, the application of (30) makes it unnecessary to perform multiple SDM calculations to find the critical pitch in a trial-and-error manner.

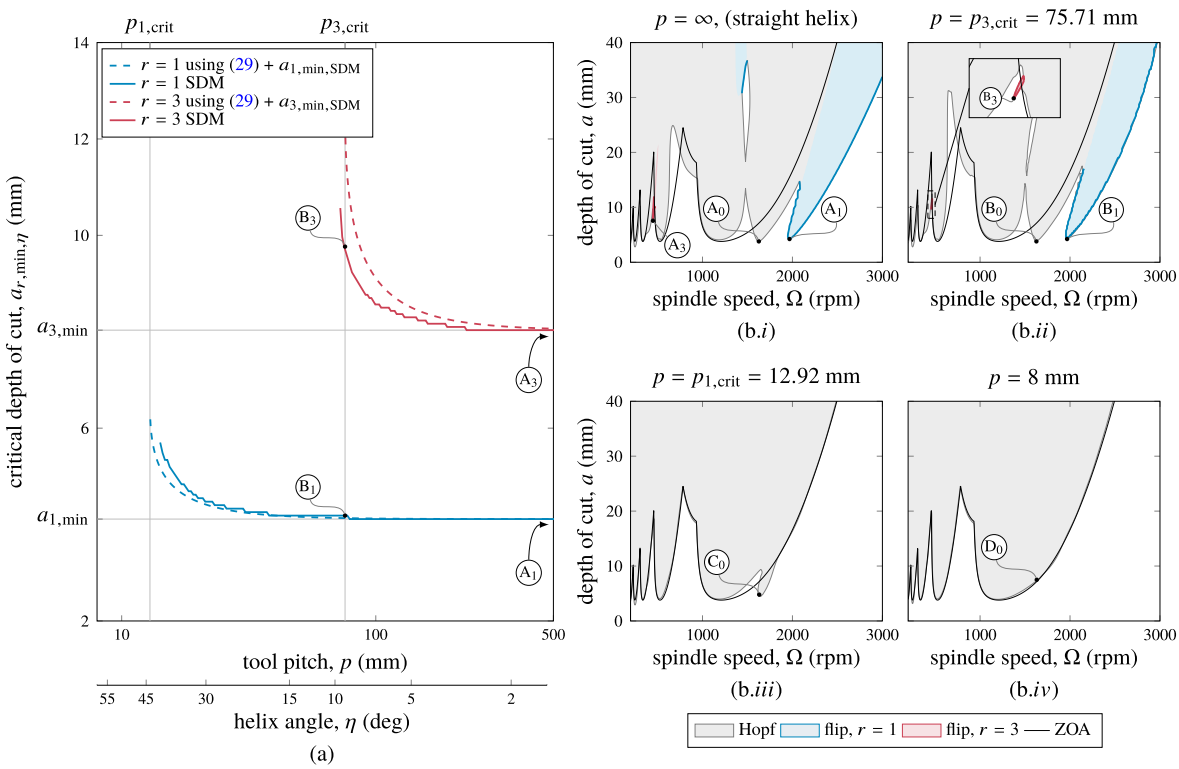

Fig. 8. Results of the numerical validation of the formula for the machine tool structure chatter case (two degree of freedom). (a) Variation of the critical flip depths of cut, calculated by the analytical formulae (29) (using the minima from semidiscretisation $a_{r,\mathrm{min,SDM}}$) and semidiscretisation. (b) Stability lobe diagrams for different values of the helix pitch. When the critical pitch of the first lobe is reached, all the flip lobes are removed. However, the effect of mode interaction is still present.

Machine tool structure chatter case

When machining at positions with large ram overhangs, machine tools usually present low frequency ram bending modes in perpendicular directions but with different frequencies that usually limit the productivity. To test the validity of the formulae over such a case, the two degree of freedom case from [3] is taken as a reference, which is inspired by the experimental measurements in a real machine with ram modes [32]. The modal parameters of the mode in x ($\theta = 0$ deg) direction are $\omega_\mathrm{n} = 45$ Hz, $\zeta = 4\%$ and $k = 30$ N/μm, while the parameters of the mode in y ($\theta = 90$ deg) direction are $\omega_\mathrm{n} = 60$ Hz, $\zeta = 4\%$ and $k = 30$ N/μm. The considered milling process is a steel ($K_\mathrm{rc} = 2000$ MPa, $K_\mathrm{rc} = 600$ MPa) roughing operation with a 16 mm diameter 4 fluted cutter and a 25% radial immersion under down-milling strategy. This special case is chosen because even though it resembles a possible industrial case, it not only shows period doubling lobes, but also important coupling (mode interaction) between the two modes happen due to the interrupted cutting, affecting the Hopf-related boundaries too. As in the previous case, the flip minimum of the straight helix case is numerically approximated by $a_{r,\mathrm{min,SDM}}$.

Fig. 8a shows the evolution of the minimums of the first two flip lobes arising in the considered spindle speed range (200–3000 rpm). The results evidence that for a two degree of freedom case, there is some deviation with respect to the numerical results even for the $r = 1$ lobe. For the $r = 3$ flip lobe, an error of 5.3% is made in the prediction of $p_{3,\mathrm{crit}}$. Consequently, as seen in Fig. 8b.*ii*, even though the area of the lobe has been much reduced, the flip island (▭) still exists in the diagram. On the contrary, for the $r = 1$ flip lobe, an error of −8.6% is made in the prediction of $p_{1,\mathrm{crit}}$. Therefore, when $p = p_{1,\mathrm{crit}}$ as in Fig. 8b.*iii* no period doubling unstable areas can be found in the stability chart. In a practical sense, the elimination of all the unstable areas related to period doubling would require minimum helix angle of 44.2°, achievable by commercial milling cutters.

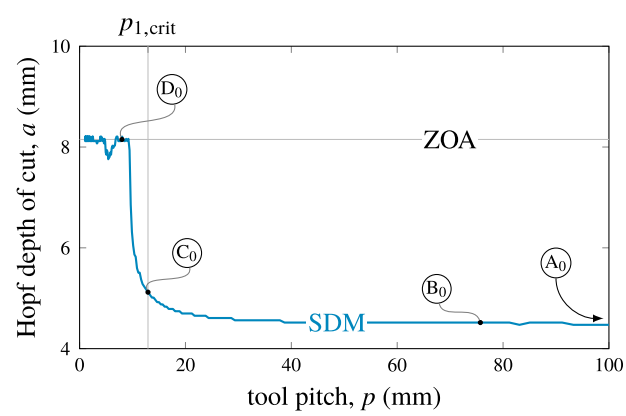

Fig. 9. Effect of decreasing the pitch helix on mode interaction. To completely remove the effect of mode interaction on Hopf boundaries, a lower helix pitch than the critical one is required for this milling case.

However, even if flip lobes have been totally canceled, slight deviation of the Hopf stability boundary can be still noticed near the critical spindle speed region ($\Omega \approx 1680$ rpm). This deviation is related to the mode interaction effect [3]. Even if this mode interaction effect is originated by the interrupted cutting as the period doubling lobes, the prediction of its Hopf boundaries is different from the flip case and therefore the effect of the helix should be different. Fig. 9 shows the evolution of the critical Hopf depth of cut in this mode interaction dominated area obtained by SDM and ZOA. The curve shows that to remove this effect and achieve a perfect match between ZOA and SDM limits in this area, a helix pitch even lower than $p_{1,\mathrm{crit}}$ is required. This is confirmed by the lobe chart in Fig. 8b.*iv*, where the effect of the

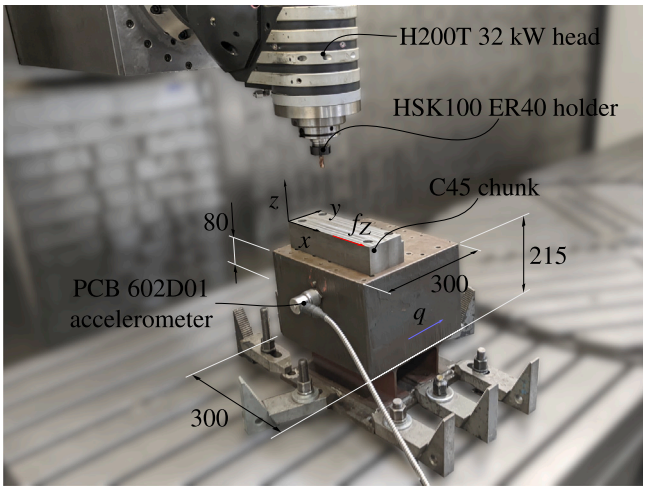

Fig. 10. Setup for the experimental cutting tests. The milling tests are performed along the x direction on a C45 chunk placed on top of a monolithic flexure that exhibits a dominant translational mode along the y direction. The cutting tests are monitored by the industrial accelerometer placed at the back of the flexure.

mode interaction has been removed. However, it must be noted that when $p = p_{1,\text{crit}}$, the deviation of Hopf boundaries produced by the mode interaction effect is already very reduced. Therefore, for most of the milling cases where the effect of mode interaction is not so evident, $p = p_{1,\text{crit}}$ should achieve a good match between ZOA and SDM boundaries.

In short, for the two degree of freedom case, the introduced formula (30) could approximate the critical pitch for each of the lobes if the critical of the straight helix pitch is provided by semidiscretisation or other numerical methods. Compared to the one degree of freedom case, for the present case the prediction of the critical for the first flip lobes slightly deviates from the actual one. Nevertheless, as in the previous cases, reaching the critical p of the first flip lobe guarantees the cancellation of all the flip lobes. Thus, (30) could be applied for an indicative or first iteration value for systems with more than one flexible cartesian directions.

4. Experimental validation

In the previous section, it has been numerically validated that for one degree of freedom systems, the introduced formulae almost exactly predict the helix pitch value that cancels the first flip lobe as well as the increase of its minimum depth of cut. For higher flip lobe orders, the accuracy decreases as more harmonics are involved. However, the prediction for the first lobe is the most relevant one, as reaching this value will cancel all the period doubling lobe orders at the same time. For higher degrees of freedom systems, the prediction of this critical pitch is less accurate, although it can provide a good indicative value to eliminate all the flip lobes too.

In this section, we experimentally validate the formulae for the first flip lobe of a one degree of freedom system, as it is the most relevant finding of the study. In this way, cutting tools with decreasing helix pitch are tested, checking which helix pitch provides absolute period doubling stability and also assessing the increase of the minimum flip depth of cut as the pitch decreases.

4.1. Experimental setup and procedure

To validate the findings of this paper concerning a one degree of freedom case, end mills with different helix pitch distances were selected, which has been confirmed to be the geometric parameter controlling the period doubling stability. As shown in Eq. (1), the helix

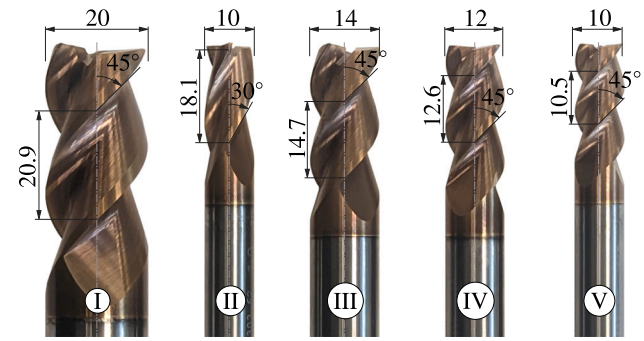

Fig. 11. End mills selected for the cutting tests. 5 carbide KENDU HMKEN end mills with different helix pitches were selected by using 4 different diameters and two sets of helix angles.

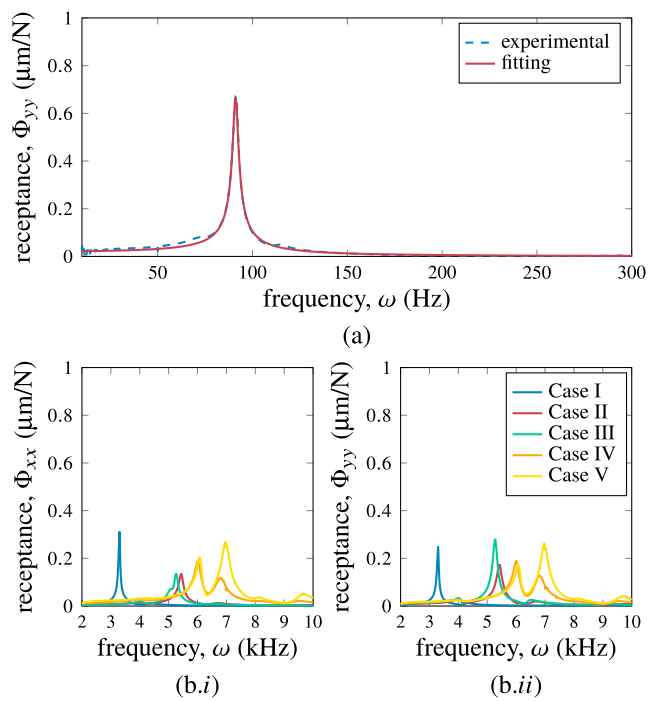

Fig. 12. Measurement of the dynamic flexibility of workpiece and cutting tools. (a) Experimental receptance and modal fitting receptance of the flexible fixture, showing the clear translational mode intended to perform the validation. (b) Experimental receptances of the 5 tools in x direction (i) and in y direction (ii). The modes presents much higher frequency and lower dynamic flexibility compared to that of the workpiece.

pitch can be adjusted by three parameters: the number of teeth Z, the helix angle η and the tool diameter D. If the dynamics remain constant, altering the number of teeth is not practical because it affects the critical depth of cut value and its corresponding spindle speed. Hence, the tool diameter and helix angle remain the parameters to control period doubling lobes.

To achieve this progressive decrease of the helix pitch, 5 carbide KENDU HMKEN end mills were selected, as illustrated in Fig. 11. To facilitate the subsequent detection of flip chatter onset, $Z = 3$ cutters were selected, as flip chatter occurs at a frequency that does not align with any of the tool rotation harmonics. From the two commercially available helix combinations, a set of 4 tools (Cases I, III, IV and V) of 45 deg helix angle (ref. 402.62) were selected with different diameters, while a 30 deg angle model (Case II, ref. 303.62) was also selected to assess the possibility of altering the helix pitch through the helix angle.

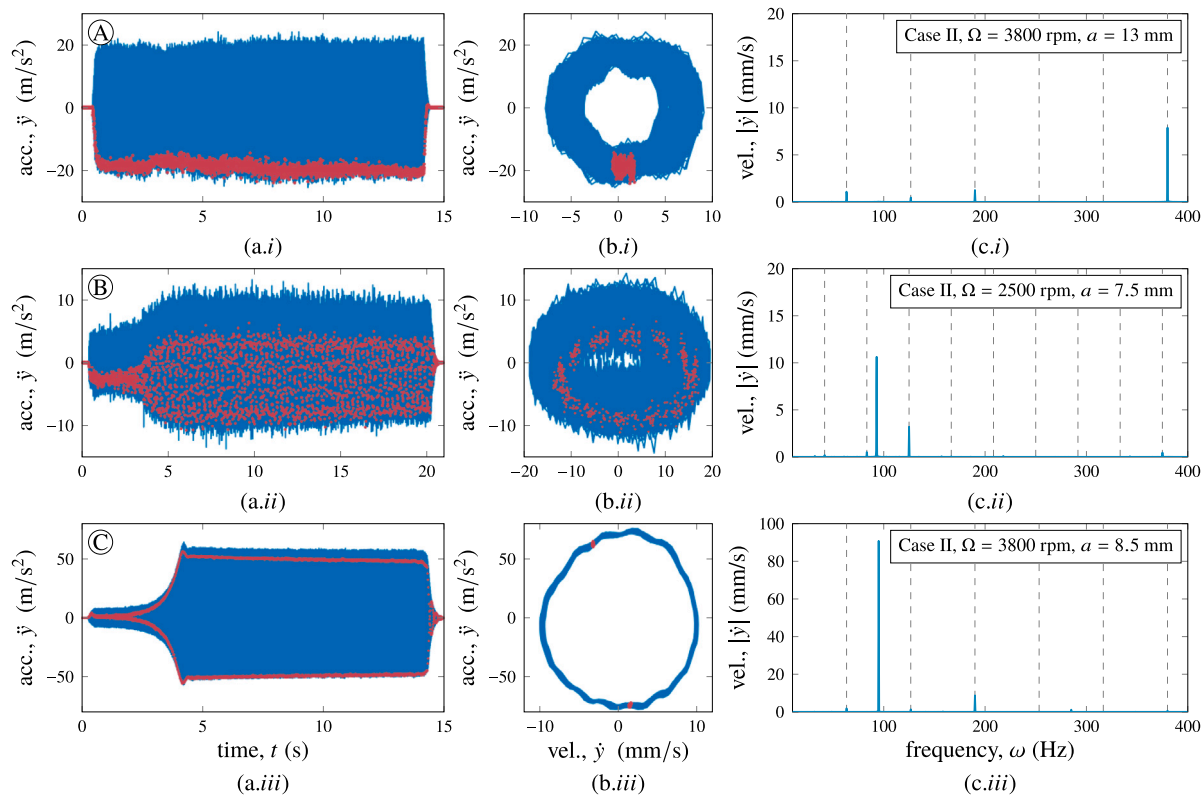

Fig. 13. Evidence and identification of period doubling chatter in the experimental tests. (a) Raw acceleration signals with one-per-tooth-passing (T_Z) sampling overlaid on them. (b) Poincare sections with one-per-tooth-passing sampling overlaid on them. (c) Root mean square velocity spectra of the accelerometer with the tool rotation harmonics. From top to bottom: cutting tests A (stable), B (Hopf unstable) and C (flip unstable) of the Case II of Fig. 15b.

Table 1
Tool geometry cases used for the cutting tests: The different combinations of cutter diameter and helix angle result in 5 different helix pitch distances.

	Milling cutter cases				
	I	II	III	IV	V
Reference	402.62.02000.20	303.62.01000	402.62.01400	402.62.01200	402.62.01000
D (mm)	20	10	14	12	10
η (deg)	45	30	45	45	45
Z	3	3	3	3	3
p (mm)	20.9	18.1	14.7	12.6	10.5
Overhang (mm)	52	32	38	38	32

Table 2
Modal, process and material parameters of the experimental tests. These parameters are then used in the analytical formulae and in the semidiscretisation calculations.

Workpiece modal parameters			
ω_n (Hz)	ζ (%)	k (N/μm)	θ (deg)
90.66 (Hz)	1.67	61.48	90
Process		Material	
f_Z (mm)	Immersion (%)	K_{rc} (MPa)	K_{rc} (MPa)
0.1	15 (down-milling)	1823.7	417.8

As shown in Table 1, this selection of tools enables a gradual reduction of the helix pitch distance controlling the stability.

The next step is to generate a single degree of freedom dynamics. In this case, the pure translational mode was sought in the workpiece side with little influence of higher frequency modes. To generate such a mode, a block mounted over parallel flexures with a C45 steel chunk placed on top of it was used (see Fig. 10). The receptance of the assembly along its flexible direction was experimentally measured by means of a PCB 086D20 impact hammer and a Dytran 3263 A accelerometer, which is shown in Fig. 12a. The modal fitting by means of dominant modal decomposition [33] resulting in the modal parameters shown in Table 2.

The cutting tests were performed in a Soraluce FMT multitasking machine with a H200T 32 kW milling head. The tools were clamped in a HSK100 ER40 toolholder. The measured receptances of the cutting tools with the overhangs in Table 1 are provided in Fig. 12b. The coefficients of the linear cutting force model were characterised by inverse mechanistic identification [34]. This was achieved by performing milling passes at different feeds per tooth (0.05, 0.1, 0.15, 0.2, 0.25 mm) on a block of the same C45 steel placed on top of a Kistler 9255B dynamometer table and performing a linear fitting of the cutting coefficients, resulting in the parameters shown in Table 2.

For each of the cutting tests, a single machining pass along the whole C45 was performed. The machining passes were performed perpendicular to the mode direction and thus $\theta = 90$ deg. The milling passes were performed under down-milling strategy with a 15% of radial engagement to attain an interrupted milling. A feed per tooth f_Z of 0.1 mm was employed for all the cutting tests.

The cutting tests were monitored using an industrial PCB 602D01 accelerometer, sampled at 8192 Hz by a Brüel and Kjaer Lan-XI model 3050 acquisition card. For each of the cutting tests, the whole pass was recorded, but only the last 4 s of the signal were analyzed to ensure the stable or unstable behavior of the point. Fig. 13a shows three examples of recorded raw acceleration signals with the one-per-tooth-passing (T_Z) sampling overlaid for an stable (Fig. 13a.i), a Hopf unstable (Fig. 13a.ii) and a period doubling unstable (Fig. 13a.iii) milling tests. The Poincaré sections obtained by the numerical integration of the acceleration signal with the one-per-tooth passing sampling are provided in Fig. 13b. To distinguish between stable and unstable cases, the velocity spectrum of the last 4 s (see Fig. 13c) was analyzed:

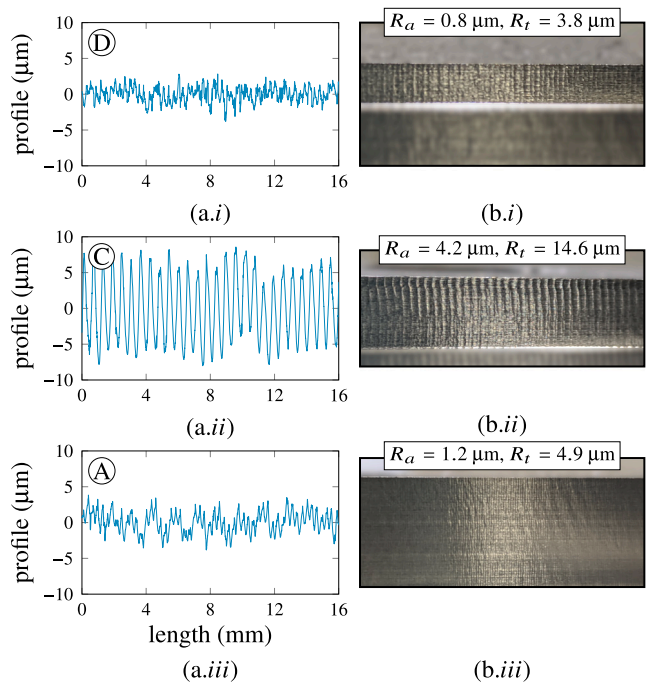

Fig. 14. Surface quality results in the experimental tests. (a) Surface profile with roughness measurements of cutting tests D (stable) and C (flip unstable) and A (stable) of the Case II of Fig. 15b. (b) Machined surface photographs of these three cutting tests with their corresponding roughness values. The measured profile, together with the roughness values and surface photographs evidence the much worse surface quality left on the part when period doubling chatter is produced, even if the axial depth of cut is lower.

if a frequency peak not coinciding with any tool rotation harmonics was detected the cutting was labeled as unstable; otherwise, it was labeled as stable. The chatter onsets were then confirmed by the one-per-tooth-passing signals and Poincaré sections (see Fig. 13ab).

The signals in Fig. 13 evidence the difference between the stable and the two types of instability of the process [35]:

- When the cutting is stable, the tooth passing and tool rotation harmonics dominate the spectrum (see Fig. 13c.*i*). If the time domain signal and Poincaré chart are observed, it shows that the flexible workpiece returns to the same position each time the tool hits the surface because of the stable cutting.
- When Hopf chatter occurs, a chatter peak arises at a frequency not coinciding with any of the tool rotation harmonics and neither with any of the analytical flip chatter distributions $\omega_c = r\omega_Z/2$. In this case, the one-per-tooth-passing sampling in Fig. 13a.*ii* and b.*ii* evidence that the chatter vibration is unsynchronised with the tooth passing as the flexible workpiece is at a different position at each tooth passing.
- When period doubling chatter occurs in the $r = 1$ lobe, a clear large-amplitude peak arises at half the tooth passing frequency (see Fig. 13c.*iii*). The onset of flip chatter is also evident in the time-domain acceleration signals and the Poincaré sections with the one-per-tooth sampling, where it can be seen that the system returns to the same position after every two tooth passings.

Fig. 14 shows the difference in surface quality between stable and period doubling unstable cases. For the stable cuttings (Fig. 14a.*i* and a.*iii*) the surface roughness is significantly inferior to the one of the flip unstable case (Fig. 14a.*ii*). As reported in [10], under stable cutting lower amplitude 0.3 mm wavelength marks dominate the surface profile due to the runout, since the profile is generated by the tooth with the maximum radial runout at each tool rotation. On the contrary, for the flip chatter case high amplitude 0.6 mm wavelength marks dominate the profile, as in this case the surface is generated by the tooth with maximum radial runout but every two tool rotations as seen in Fig. 13a.*iii* and b.*iii*.

4.2. Results

In the cutting tests two findings were validated. On the one hand, the main objective was to demonstrate that if the helix pitch is below the critical $p_{1,\mathrm{crit}}$, the period doubling lobe – along with all higher-order flip lobes – disappear from the stability diagram. On the other hand, as the helix pitch decreases and approaches $p_{1,\mathrm{crit}}$, the sudden increase of the minimum flip depth of cut $a_{1,\min,\eta}$ was also validated.

For the present dynamics, number of teeth and radial engagement, the analytical minimum flip depth of cut calculated with (24) is 3.62 mm. Therefore, for the first ($r = 1$) flip lobe the critical pitch given by (30) that would totally vanish the flip islands results in 11.37 mm. Thus, for the combination of cutting tools selected for the validation, the flip island should vanish for the Case V, while flip chatter should still be possible for the rest of the cutters.

To validate this, cutting tests were performed for each of the cases within and around the period doubling islands, guided by the stability lobe diagrams (and chatter frequency diagrams) obtained through semidiscretisation for the five cases described in Table 1, as shown in Fig. 15a–e. For each case, the experimental minimum flip depth of cut and the corresponding critical spindle speed were determined by identifying the lowest unstable depth of cut in the region. The experimental cutting tests at these critical spindle speeds for each of the cases are then compared to the theoretical critical depth of cut curve provided by (29) in Fig. 15f.

The results confirm the main finding of this paper: as seen in Fig. 15a–d, when the pitch is above the critical value $p_{1,\mathrm{crit}}$ provided by (30), $r = 1$ period doubling lobe still exists and period doubling chatter can be experimentally attained. On the contrary, once the helix pitch falls below the critical value the $r = 1$, period doubling lobe is totally removed (see Case V in Fig. 15e). Consequently, all higher-order period doubling lobes will also vanish from the stability diagram, as seen in Section 3.1.

This demonstrates that a robust design of cutter helix can be achieved by simply estimating the minimum depth of cut of the straight fluted case, which can be determined either through frequency domain methods, discrete-time methods, or by utilising the analytical formula in (24) in the case of one degree of freedom systems. As previously mentioned, if the proper helix pitch p is selected, it is guaranteed that the interrupted cutting will not lead to flip instability and that the fast ZOA solution will perfectly match the solutions provided by more time-consuming discrete-time algorithms.

Furthermore, the experimental results for the critical depth of cut shown in Fig. 15(f) confirm that as the helix pitch approaches the critical value, the minimum period doubling depth of cut increases. However, as observed in the numerical validation conducted in the previous section, this increase is significant only for helix pitch values very close to the threshold at which the flip lobes disappear. That is, the flip minimum will only undergo significant variations when the flip lobe is very close to be totally removed by the helix.

5. Conclusions

This paper has introduced a formula for selecting the cutter helix pitch to eliminate the likelihood of period doubling chatter in milling processes. By performing harmonic truncation, a closed-form formula that relates the minimum rth period doubling depth of cut of a helical end mill to that of a straight helix cutter was derived. This formula predicts that if the helix pitch is below the critical depth of cut of the straight helix cutter multiplied by π and the flip lobe order r, the flip lobe will totally vanish. This means that if the first flip lobe is

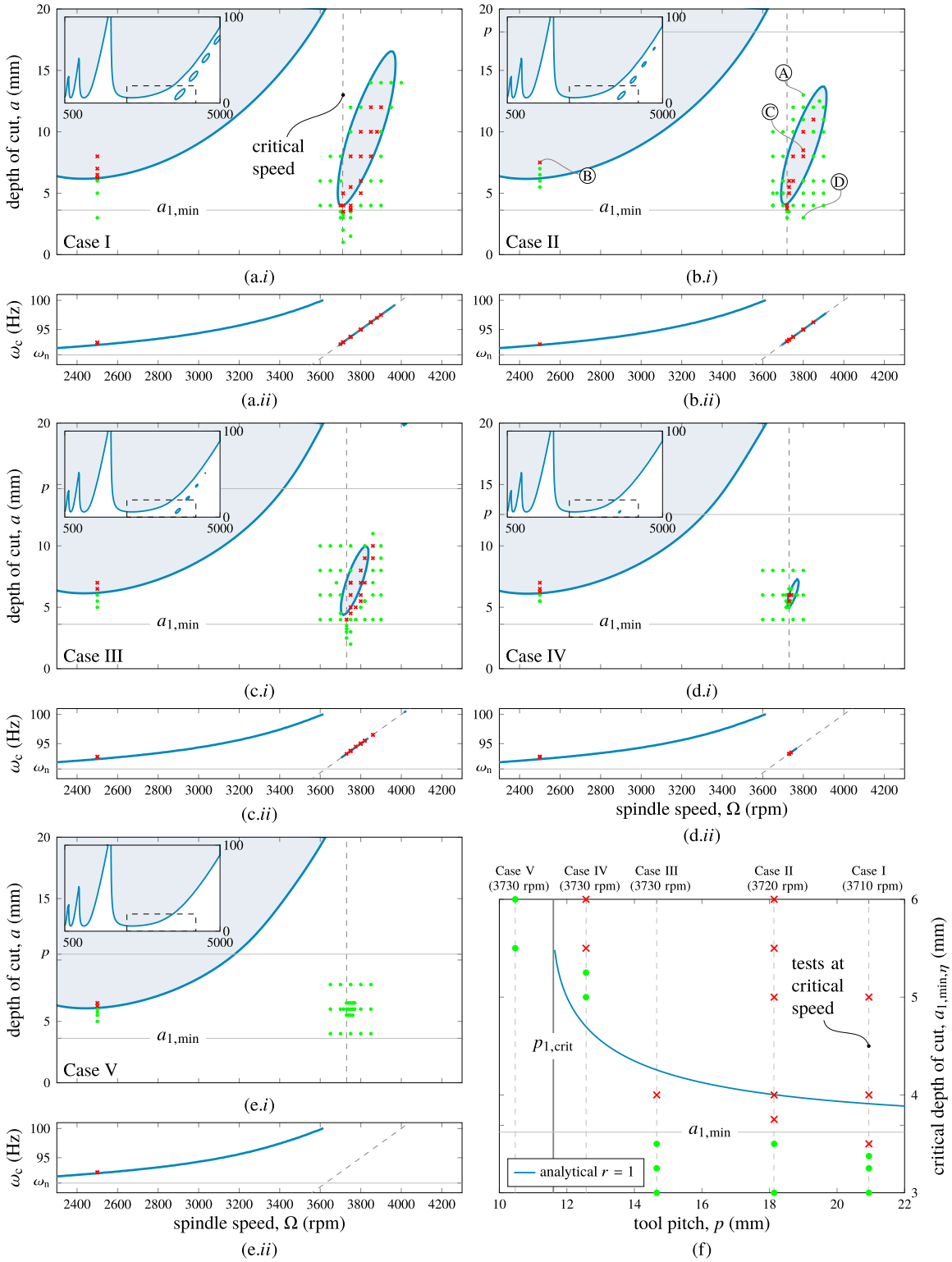

Fig. 15. Results on the experimental cutting tests for cutters with different pitch. (a)–(e) Stability lobe diagrams with chatter frequency diagrams obtained by SDM for the milling tools I - V in Table 1 with the results of the experimental cutting tests overlaid on them. (f) Helix pitch — critical depth of cut curves obtained analytically from (29) together with the experimental critical depths of cut extracted from the stability lobes at their corresponding critical spindle speeds.

removed by selecting a helix pitch equal to the critical, all the flip lobes, including the higher order ones will be removed.

The validity of the formula was tested both numerically through semidiscretisation calculations and experimental cutting tests. For one degree of freedom systems, this rule provides almost exact prediction for the first flip lobe, vanishing not only the first flip lobe but also all the higher order ones too. In the case of multiple degree of freedom systems, it still offers a good indicative value for flip avoidance, although the minimum of the straight fluted cutter has to be numerically calculated in this case.

The formula presented in this paper offers valuable guidance for selecting the optimal tool helix to completely eliminate period doubling chatter. By estimating the minimum flip depth of cut by means of the analytical formula or by classic numerical stability algorithms, the diameter and helix angle combination that guarantees a flip chatter-free machining regardless of the considered cutting parameters can be selected. In addition, the critical pitch value establishes a boundary to decide when to use the zeroth order algorithm or discrete-time methods for stability prediction: if the helix pitch remains below the value provided by the formula, the zeroth order algorithm and discrete-time methods will yield the same stability limits, making the zeroth order algorithm the preferred choice due to its lower computational effort. On the contrary, if the helix pitch exceeds the critical value, deviations in the stability limits should be expected due to period doubling lobes, and therefore discrete-time algorithms should be applied to accurately predict the stability of the system.

CRediT authorship contribution statement

M. Sanz-Calle: Writing – original draft, Validation, Software, Investigation, Conceptualization. **A. Iglesias:** Writing – review & editing, Supervision, Project administration. **L.N. López de Lacalle:** Supervision, Resources. **Z. Dombovari:** Writing – review & editing, Conceptualization. **J. Munoa:** Writing – review & editing, Supervision, Investigation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

Acknowledgments

This research has been funded by ROBOCOMP (Plan Tecnológico Aeronáutico. Ministerio de Ciencia, Innovación y Universidades. Grant number: EXP 00152596/PTAG-20221002) and MITIGA (Ministerio de Ciencia, Innovación y Universidades. Fondos NextGenEU. Grant number: CPP2022-009918) projects.

References

[1] J. Munoa, X. Beudaert, Z. Dombovari, Y. Altintas, E. Budak, C. Brecher, et al., Chatter suppression techniques in metal cutting, CIRP Ann. - Manuf. Technol. 65 (2) (2016) 785–808, http://dx.doi.org/10.1016/j.cirp.2016.06.004.

[2] Y. Altintas, E. Budak, Analytical prediction of stability lobes in milling, CIRP Ann 44 (1) (1995) 357–362, http://dx.doi.org/10.1016/S0007-8506(07)62342-7.

[3] J. Munoa, Z. Dombovari, I. Mancisidor, Y. Yang, M. Zatarain, Interaction between multiple modes in milling processes, Mach. Sci. Technol. 17 (2) (2013) 165–180, http://dx.doi.org/10.1080/10910344.2012.747935.

[4] M.A. Davies, J.R. Pratt, B.S. Dutterer, T.J. Burns, Stability of low radial immersion milling, CIRP Ann. - Manuf. Technol. 49 (1) (2000) 37–40, http://dx.doi.org/10.1016/S0007-8506(07)62891-1.

[5] I. Minis, R. Yanushevsky, A new theoretical approach for the prediction of machine tool chatter in milling, Trans. ASME, J. Manuf. Sci. Eng. 115 (1) (1993) 1–8, http://dx.doi.org/10.1115/1.2901633.

[6] M. Sanz-Calle, J. Munoa, A. Iglesias, L. López de Lacalle, Z. Dombovari, Semianalytic stability model for interrupted milling, Int. J. Mach. Tools Manuf. 187 (2023) 104005, http://dx.doi.org/10.1016/j.ijmachtools.2023.104005.

[7] T. Insperger, G. Stepan, Semi-Discretization for Time-Delay Systems, Vol. 178, Springer-Verlag, New York, ISBN: 9781461403357, 2011.

[8] Y. Ding, L.M. Zhu, X.J. Zhang, H. Ding, A full-discretization method for prediction of milling stability, Int. J. Mach. Tools Manuf. 50 (5) (2010) 502–509, http://dx.doi.org/10.1016/j.ijmachtools.2010.01.003.

[9] D. Bachrathy, T. Insperger, G. Stépán, Surface properties of the machined workpiece for helical mills, Mach. Sci. Technol. 13 (2) (2009) 227–245, http://dx.doi.org/10.1080/10910340903012167.

[10] S. Seguy, T. Insperger, L. Arnaud, G. Dessein, G. Peigné, Suppression of period doubling chatter in high-speed milling by spindle speed variation, Mach. Sci. Technol. 15 (2) (2011) 153–171, http://dx.doi.org/10.1080/10910344.2011.579796.

[11] M. Zatarain, J. Muñoa, G. Peigne, T. Insperger, Analysis of the influence of mill helix angle on chatter stability, CIRP Ann. - Manuf. Technol. 55 (1) (2006) 365–368, http://dx.doi.org/10.1016/S0007-8506(07)60436-3.

[12] T. Insperger, J. Munoa, M.a. Zatarain, G. Peigné, Unstable islands in the stability chart of milling processes due to the helix angle, in: CIRP 2nd International Conference on High Performance Cutting, Vancouver, Canada, 2006, pp. 12–13.

[13] K. Zhou, J. Zhang, C. Xu, P. Feng, Z. Wu, Effects of helix angle and multi-mode on the milling stability prediction using full-discretization method, Precis. Eng. 54 (2018) 39–50, http://dx.doi.org/10.1016/j.precisioneng.2018.04.016.

[14] B. Patel, B. Mann, K. Young, Uncharted islands of chatter instability in milling, Int. J. Mach. Tools Manuf. 48 (1) (2008) 124–134, http://dx.doi.org/10.1016/j.ijmachtools.2007.06.009.

[15] Z. Dombovari, R. Toth, A. Iglesias, D. Bachrathy, G. Stepan, Flip-validated milling process in hardware-in-the-loop environment, CIRP Ann 72 (1) (2023) 369–372, http://dx.doi.org/10.1016/j.cirp.2023.04.089.

[16] J. Munoa, Desarrollo de un modelo general para la predicción de la estabilidad del proceso de fresado. Aplicación al fresado periférico, al planeado convencional y a la caracterización de la estabilidad dinámica de fresadoras universales, (Ph.D. thesis), Mondragon University, 2007.

[17] F.A. Khasawneh, O.A. Bobrenkov, B.P. Mann, E.A. Butcher, Investigation of period-doubling islands in milling with simultaneously engaged helical flutes, J. Vib. Acoust. 134 (2) (2012) 021008, http://dx.doi.org/10.1115/1.4005022.

[18] A. Iglesias, Z. Dombovari, G. Gonzalez, J. Munoa, G. Stepan, Optimum selection of variable pitch for chatter suppression in face milling operations, Materials 12 (1) (2019) http://dx.doi.org/10.3390/ma12010112.

[19] S. Jiang, D. Zhan, Y. Liu, Y. Sun, J. Xu, Modeling of variable-pitch/helix milling system considering axially varying dynamics with cutter runout offset and tilt effects, Mech. Syst. Signal Process. 168 (2022) 108674, http://dx.doi.org/10.1016/j.ymssp.2021.108674.

[20] M. Ding, X. Liu, C. Yue, S. Sun, W. Xia, Research on design and milling performance of unequal gradient helix and variable pitch end mill, J. Manuf. Process. 106 (2023) 347–362, http://dx.doi.org/10.1016/j.jmapro.2023.10.013.

[21] J. Niu, Y. Ding, L. Zhu, H. Ding, Mechanics and multi-regenerative stability of variable pitch and variable helix milling tools considering runout, Int. J. Mach. Tools Manuf. 123 (2017) 129–145, http://dx.doi.org/10.1016/j.ijmachtools.2017.08.006.

[22] A.R. Yusoff, N.D. Sims, S. Turner, Experimental validation of chatter stability for variable helix milling tools, IOP Conf. Ser.: Mater. Sci. Eng. 26 (1) (2011) http://dx.doi.org/10.1088/1757-899X/26/1/012010.

[23] J. Niu, Y. Ding, Z. Geng, L. Zhu, H. Ding, Patterns of regenerative milling chatter under joint influences of cutting parameters, tool geometries, and runout, J. Manuf. Sci. Eng. 140 (12) (2018) 121004, http://dx.doi.org/10.1115/1.4041250.

[24] T. Hayasaka, A. Ito, E. Shamoto, Generalized design method of highly-varied-helix end mills for suppression of regenerative chatter in peripheral milling, Precis. Eng. 48 (2017) 45–59, http://dx.doi.org/10.1016/j.precisioneng.2016.11.004.

[25] R. Szalai, G. Stépán, S.J. Hogan, Global dynamics of low immersion high-speed milling, Chaos 14 (4) (2004) 1069–1077, http://dx.doi.org/10.1063/1.1807395.

[26] W.T. Corpus, W.J. Endres, Added stability lobes in machining processes that exhibit periodic time variation, part 1: An analytical solution, Trans. ASME, J. Manuf. Sci. Eng. 126 (3) (2004) 467–474, http://dx.doi.org/10.1115/1.1765137.

[27] A. Iglesias, J. Munoa, J. Ciurana, Z. Dombovari, G. Stepan, Analytical expressions for chatter analysis in milling operations with one dominant mode, J. Sound Vib. 375 (2016) 403–421, http://dx.doi.org/10.1016/j.jsv.2016.04.015.

[28] E. Budak, Y. Altintas, Analytical prediction of chatter stability in milling—part I: General formulation, J. Dyn. Syst. Meas. Control Trans. ASME 120 (1) (1998) 22–30, http://dx.doi.org/10.1115/1.2801317.

[29] Z. Dombovari, A. Iglesias, M. Zatarain, T. Insperger, Prediction of multiple dominant chatter frequencies in milling processes, Int. J. Mach. Tools Manuf. 51 (6) (2011) 457–464, http://dx.doi.org/10.1016/j.ijmachtools.2011.02.002.

[30] J. Munoa, M. Sanz-Calle, Z. Dombovari, A. Iglesias, J. Pena-Barrio, G. Stepan, Tuneable clamping table for chatter avoidance in thin-walled part milling, CIRP Ann 00 (2020) 14–17, http://dx.doi.org/10.1016/j.cirp.2020.04.081.

[31] I. Mancisidor, A. Urkiola, R. Barcena, J. Munoa, Z. Dombovari, M. Zatarain, Receptance coupling for tool point dynamic prediction by fixed boundaries approach, Int. J. Mach. Tools Manuf. 78 (2014) 18–29, http://dx.doi.org/10.1016/j.ijmachtools.2013.12.002.

[32] O. Franco, X. Beudaert, A. Iglesias, Z. Dombovari, K. Erkorkmaz, J. Munoa, Optimal cutting condition selection for high quality receptance measurements by sweep milling force excitation, Int. J. Mach. Tools Manuf. 176 (2022) 103873, http://dx.doi.org/10.1016/j.ijmachtools.2022.103873.

[33] Z. Dombovari, Dominant modal decomposition method, J. Sound Vib. 392 (2017) 56–69, http://dx.doi.org/10.1016/j.jsv.2016.12.012.

[34] Y. Altintas, Manufacturing Automation: Metal Cutting Mechanics, Machine Tool Vibrations, and CNC Design, second ed., Cambridge University Press, 2012, http://dx.doi.org/10.1017/CBO9780511843723.

[35] B.P. Mann, T. Insperger, P.V. Bayly, G. Stépán, Stability of up-milling and down-milling, part 2: Experimental verification, Int. J. Mach. Tools Manuf. 43 (1) (2003) 35–40, http://dx.doi.org/10.1016/S0890-6955(02)00160-8.